Phage Therapy for MDR/XDR/PDR *Pseudomonas aeruginosa*: A Systematic Review and Meta-Analysis of Proportions

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Abstract

Background: Multidrug-resistant (MDR), extensively drug-resistant (XDR), and pandrug-resistant (PDR) *Pseudomonas aeruginosa* is a WHO priority pathogen with few options. Phage therapy has expanded from compassionate use to early-phase trials without resistance- or route-stratified synthesis. Methods: We conducted a PRISMA 2020 meta-analysis of proportions (binomial-normal GLMM), pooling clinical success, safety, eradication, and mortality across phage-therapy studies of MDR/XDR/PDR *P. aeruginosa*, stratified by resistance category, route, and modality. Results: Thirty-one pooling-eligible study-arms (25 studies, 82 patients) yielded clinical success in 76.1% (95% CI 64.2–84.9%), a proportion of the published record rather than of treated patients. Of 36 pre-specified strata, 13 met threshold, including MDR clinical success 73.0% (95% CI 55.0–85.7%). XDR, PDR, every specific route, and monotherapy remained below threshold. All 13 pooled cells are GRADE Very low certainty by construction, and model choice moves most estimates by more than five percentage points. Conclusions: These proportions should not inform a treatment decision; the review's usable output is a map of which comparisons the published record cannot yet support.

Keywords: Bacteriophages; Pseudomonas aeruginosa; Drug Resistance, Multiple, Bacterial; Phage Therapy; Systematic Reviews as Topic; Meta-Analysis as Topic

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1 Introduction

Pseudomonas aeruginosa appears on the World Health Organization's priority list of antibiotic-resistant pathogens because of its capacity to accumulate resistance to nearly every class of antipseudomonal agent (Tacconelli et al., 2018). That framing needs one correction, and it cuts against rather than for the urgency of this review's question: the 2024 update to the priority list *downgraded* carbapenem-resistant *P. aeruginosa* from Critical to High (World Health Organization, 2024). We note it explicitly rather than quietly citing the superseded 2018 list. The organism remains a High-priority pathogen for which salvage options are genuinely scarce, which is sufficient motivation; overstating its rank is not necessary and would not survive review.

A second correction is owed to the premise itself. It is no longer accurate to say the antipseudomonal pipeline has stalled: ceftolozane–tazobactam, ceftazidime–avibactam, imipenem–relebactam and cefiderocol all reached clinical use in the last decade and between them recovered activity against many isolates that older definitions call multidrug-resistant. The scarcity this review addresses is therefore not an absence of new agents but the narrower and harder situation those agents leave behind — the patient in whom they have also failed, or cannot be used. That distinction has a formal counterpart. The Magiorakos categories (MDR/XDR/PDR) count how many antimicrobial *categories* an isolate resists, a measure the new agents have made less decision-relevant; Kadri et al. (2018) instead define *difficult-to-treat resistance* (DTR) as non-susceptibility to *all* first-line agents — every β-lactam and every fluoroquinolone — which is the point at which a clinician is forced onto reserve therapy. DTR, not MDR, is the category current guidance uses for treatment decisions in *P. aeruginosa*. We retain the Magiorakos axis as primary because it is what the published literature reports, and we carry DTR as a separate stratum precisely to show what that reporting practice costs (Sections 2.2 and 3.4). Once an isolate meets criteria for multidrug resistance (MDR), extensive drug resistance (XDR), or pandrug resistance (PDR) under the Magiorakos framework (Magiorakos et al., 2012), treatment options narrow quickly. The antipseudomonal development pipeline has not kept pace, and bacteriophage therapy—administered alone or alongside antibiotics—has moved from isolated compassionate-use reports toward a small but growing set of early-phase trials as a salvage option for infections that no longer respond to conventional regimens.

This evidence base has already been synthesized many times. Since El Haddad et al. (2019) reviewed phage therapy against ESKAPE pathogens in 2019, at least eight further systematic reviews, meta-analyses, and scoping reviews of human phage therapy have appeared, including a large multi-author synthesis in *Lancet Infectious Diseases* (Uyttebroek et al., 2022) and, most recently, an individual-patient-data meta-analysis of 130 studies spanning multiple pathogens (Liu, Thong, et al., 2025). None restricts its pooling to *P. aeruginosa* specifically, and none stratifies pooled outcomes by resistance category or by route of phage administration—the two variables clinicians most need resolved when choosing a regimen for a given patient and infection site.

Our original aim was narrower still: to compare, within the same studies, phage-antibiotic combination therapy against antibiotic monotherapy for MDR/XDR/PDR *P. aeruginosa* and to estimate a pooled effect size for clinical success. A systematic search of PubMed/MEDLINE and ClinicalTrials.gov found no study reporting both a genuine antibiotic-monotherapy arm and a phage-antibiotic combination arm in a *P. aeruginosa*-specific, MDR/XDR/PDR-defined population, so this comparison could not be constructed under any available within-study design. We therefore reframed the review around pooled proportions rather than a paired effect estimate: the proportion of patients achieving clinical success and the proportion experiencing a safety event across the single-arm and comparative studies that do exist, with pre-specified stratification by resistance category (MDR, XDR, or PDR), route of administration, and treatment modality (antibiotic monotherapy vs. phage-antibiotic combination).

Specifically, among patients with MDR, XDR, or PDR *P. aeruginosa* infections treated with bacteriophage therapy, we asked: what proportion achieve clinical success, what proportion experience an adverse event, and do these proportions differ by resistance category, administration route, or treatment modality?

We found the current evidence base able to answer only part of the stratified question it was designed to ask. Pooling across all included studies yielded a clinical-success proportion of 76.1% (95% CI 64.2–84.9%), directionally consistent with the high success rates reported by prior, non-stratified syntheses—but this figure comes from a compassionate-use population selected for refractory infection after antibiotic failure and should not be read as evidence that phage therapy outperforms, or underperforms, any comparator. Stratification succeeded, but only just, for one resistance class: MDR isolates specifically now have a pooled estimate (73.0% clinical success, 95% CI 55.0–85.7%). We present this as a single-cohort characterization rather than a headline achievement: it draws two-thirds of its patients from one cohort and clears the pre-specified evidentiary threshold only under author-reported resistance labels—restricted to independently re-derived classifications, the MDR cell collapses to three studies and four patients and cannot be pooled at all. XDR, PDR, every specific administration route, and modality all fell short of the number of studies and patients needed for a stable pooled estimate, and despite a targeted search, no eligible antibiotic-monotherapy arm was identified—so the MDR-versus-XDR comparison that motivated this review remains unanswered even though MDR alone is not. Documenting exactly which parts of this stratified question the current evidence can and cannot answer, rather than forcing an estimate the data cannot support or dismissing the one part that succeeded, is the contribution of this review.

2 Methods

This review follows the PRISMA 2020 reporting standard (Page et al., 2021) and the methodological guidance of the Cochrane Handbook (Higgins et al., 2023), adapted throughout for a synthesis of proportions rather than a synthesis of comparative treatment effects, for the reasons given in the Introduction. Every design choice below—eligibility, search, extraction, risk-of-bias assessment, and the pooling model itself—was fixed before formal data extraction began, and is reported here exactly as specified, including the places where the evidence base did not cooperate with the plan and the places where this review's own execution fell short of its own plan.

2.1 Protocol and Registration

This review was not registered in PROSPERO before literature screening or data extraction began. The design pivoted once during discovery, from a comparative-effect meta-analysis (phage-antibiotic combination versus antibiotic monotherapy) to the stratified meta-analysis of proportions reported here, after two independent search rounds found zero to one qualifying dual-arm comparative studies for MDR/XDR *P. aeruginosa.* That pivot needed to happen before a protocol could be written that the team would not have to immediately amend. That scoping work included four rounds of literature searching (Section 2.3) and one informal, single-reviewer screening pass of a 201-document local PDF corpus, conducted against this review's PICO criteria but without the dual-independent review PROSPERO registration presumes. PROSPERO's own policy does not prohibit preliminary scoping before registration; it prohibits registering a protocol once data extraction and synthesis are already substantively underway. By that standard, registration remained available through the design-pivot and literature-search stages of this project, and did not occur. Data extraction and the statistical synthesis reported in Results proceeded before a protocol was ever submitted – a genuine deviation from prospective registration – and that extraction itself was conducted by a single reviewer rather than the dual-independent process a registered protocol would specify (Sections 2.4 and 2.5). The full scoping timeline is as follows: two PubMed/ClinicalTrials.gov search rounds and a targeted third and fourth round (Section 2.3), one ad hoc single-reviewer PDF screen, and the resulting design pivot, all preceded any protocol filing, and no protocol was filed before extraction and pooling were carried out.

We recommend retrospective PROSPERO registration disclosing this exact sequence before this review is submitted for publication.

2.2 Eligibility Criteria

Eligibility follows a PICO framework, broadened partway through discovery once the design pivoted from a comparative-effect estimand to a proportion estimand (Section 2.1): under the original comparative-only design, a study needed both a phage-containing arm and a within-study antibiotic-monotherapy arm to qualify, a requirement that essentially no study in this literature satisfies. Under the current design, single-arm and comparative studies are equally eligible, because every outcome is computed at the study-arm level rather than as a paired comparison.

- **Population.** Patients with an infection attributed to multidrug-resistant (MDR), extensively drug-resistant (XDR), or pandrug-resistant (PDR) *Pseudomonas aeruginosa*, classified per the Magiorakos et al. (2012) criteria where the primary study reports sufficient susceptibility data, and per the study's own MDR/XDR/PDR label otherwise (Section 2.5). Three cases must be distinguished, and this review treats them differently.

  First, where a study reports susceptibility data that *positively fail* the Magiorakos threshold, the arm does not meet the population criterion and is excluded, whatever the study's own descriptive language. Second, and consequentially, an arm is excluded where its source population was *recruited without any resistance requirement* — that is, where a trial's own eligibility criteria admit patients on the basis of *P. aeruginosa* infection or chronic airway colonization alone. For such arms the missing resistance label is not a reporting gap: the population demonstrably was not selected for resistance, and pooling it would place patients who may be fully susceptible inside a review of resistant infection. Third, where an individual compassionate-use or salvage report simply does not *report* a susceptibility panel, the arm is retained and coded *not classifiable*, because absence of a reported panel is not evidence of susceptibility and the population is the refractory, antibiotic-failed population this review targets.

The distinction between the second and third cases is the load-bearing one, and we state it plainly: a missing susceptibility panel in a salvage case report is a documentation gap, whereas a missing resistance criterion in a trial protocol is a different population. Applying the second rule consistently removes every randomized trial from the synthesis — PhagoBurn, whose burn-wound population required no MDR status; and the BX004-A and SWARM-P.a./AP-PA02 cystic-fibrosis trials, which enrolled on chronic pulmonary colonization — together with three further completed trials identified in a registry sweep (Section 2.3). We accept that cost deliberately. Those trials were never estimating this review's estimand, and they are more informative as external comparative evidence (Section 4.2) than as single-arm proportions inside a synthesis whose population they do not share. What remains is a corpus of compassionate-use and salvage reports in which resistance is established for all but two arms, and a sensitivity analysis confirms those two move no pooled estimate by more than a few percentage points (Section 3.6).

- **Resistance classification and the DTR axis.** Arms are classified on the Magiorakos et al. (2012) axis (MDR/XDR/PDR/not classifiable) as above. A *fourth* stratification axis records difficult-to-treat resistance (Kadri et al., 2018), defined as non-susceptibility to all β-lactams and all fluoroquinolones. DTR is not a fifth Magiorakos level and is not nested inside that axis: it asks a different question — whether every first-line agent is lost — so an isolate can be MDR and DTR, or MDR and *not* DTR while cefepime remains active. It is therefore stratified in parallel. Coding is deliberately conservative: an arm is DTR-positive only where PDR status *entails* it, since non-susceptibility to all agents in all categories necessarily includes all β-lactams and both fluoroquinolones; every other arm is recorded as *not derivable* rather than assumed negative, because XDR permits retained susceptibility in up to two categories and MDR permits far more. Magiorakos is retained as the primary axis for one reason, stated plainly: it is what this literature publishes. The DTR axis is carried not because we expect to estimate it but because the pattern of its failure is itself the finding (Section 3.4).

- **Intervention.** Bacteriophage therapy, administered as monotherapy or adjunctively with antibiotics, by any route.

- **Comparator.** None required. Where a genuine within-study comparator exists (an antibiotic-only or placebo arm), it is retained and reported, but its presence is not an eligibility condition.

- **Outcomes.** Clinical success and any adverse event (primary); microbiological eradication, mortality, length of hospitalization, and resistance emergence during treatment (secondary), each defined per the primary study's own reporting given the expected heterogeneity in how "success" is defined across this literature. Because each study defines clinical success on its own terms *and* across non-comparable infection syndromes — biliary, cystic-fibrosis lung, prosthetic-joint, vascular-graft, urinary-tract, bacteremia, and osteomyelitis among those represented in this corpus — the pooled clinical-success proportion aggregates outcomes that are not the same clinical construct, and should be read as a rough descriptive summary rather than a harmonized effect estimate. A syndrome-harmonized success definition was not achievable given the heterogeneous reporting of the primary literature, and this constraint is carried forward as a limitation (Section 4.4).

- **Study designs.** Randomized controlled trials, prospective and retrospective cohorts, case series, and case reports, including compassionate-use reports.

A study-arm was excluded if its causative pathogen could not be separated from a mixed-pathogen cohort (coded `mixed-not-separable` in the extraction schema; Section 2.5), regardless of how well the rest of the study otherwise matched. The primary electronic search (Section 2.3) was restricted to records published from 2016 onward across all three core databases. This 2016 cutoff was chosen as a pre-specified window that comfortably predates the modern clinical phage-therapy case-report era for *P. aeruginosa* — which begins in earnest around 2018–2019 (the earliest included study is Duplessis et al., 2018) — while still excluding the sparse, mostly single-institution reports from before the current compassionate-use framework. A targeted screen of the 2016–2018 window (Section 2.3) confirmed that the pre-2019 literature for this pathogen is almost entirely preclinical (phage isolation, characterization, biofilm and animal-model work); the only extractable human clinical studies in that window are Duplessis et al. (2018) and the PhagoBurn trial (Jault et al., 2019), both of which are included. A small number of still-older studies were also identified through citation-chasing of other reviews' reference lists (Section 2.3) and screened against the same eligibility criteria irrespective of publication date; one such study (Rhoads et al., 2009) failed eligibility on grounds unrelated to date and is retained only as background context. The discovery-phase search separately confirmed that, contrary to the project's initial premise, at least eight further systematic reviews, meta-analyses, or scoping reviews of human phage therapy have appeared since El Haddad et al. (2019), including a broader synthesis in a comparably high-impact journal (Uyttebroek et al., 2022) and a 130-study individual-patient-data meta-analysis published only months before this review's search began (Liu, Thong, et al., 2025).

2.3 Information Sources and Search Strategy

The primary electronic search was run directly against PubMed/MEDLINE using the NCBI E-Utilities API (`esearch`, `esummary`, `efetch`), and against ClinicalTrials.gov using its structured API (v2), across four documented rounds between 2026-07-18 and 2026-07-20. A representative query, run to test the central negative claim of this review (that essentially no dual-arm comparative study of phage-antibiotic combination versus antibiotic monotherapy exists for MDR/XDR *P. aeruginosa*), took the form:

```
("Pseudomonas aeruginosa"[tiab] OR "P. aeruginosa"[tiab]) AND ("phage"[tiab] OR
"bacteriophage"[tiab]) AND ("multidrug-resistant"[tiab] OR "MDR"[tiab] OR "extensively
drug-resistant"[tiab] OR "XDR"[tiab]) AND ("matched cohort"[tiab] OR "propensity
score"[tiab] OR "registry"[tiab] OR "retrospective comparative"[tiab] OR "case-control"[tiab])
```

This query, built directly from the terms a referee would be expected to demand, returned one hit, and that hit was a conference-abstract index record with no substantive connection to phage therapy: an effective true hit count of zero. Across all four rounds, PubMed queries of this kind returned 54 raw hits screened individually for the comparative-design question and 48 raw hits for the systematic-review/meta-analysis recheck; ClinicalTrials.gov structured queries returned 10 additional trial records screened individually. Round 3 added five previously uncatalogued comparative or randomized trials (Krakhotkin et al., 2025; Leitner et al., 2021; Karn et al., 2024; Rhoads et al., 2009; Stanley et al., 2025), found by citation-chasing the risk-of-bias tables in the largest existing phage-therapy meta-analysis's supplementary dataset (Liu, Thong, et al., 2025) instead of a new keyword search. This is standard, low-risk PRISMA practice: only citation leads were imported at this stage (Section 2.5 details the boundary between this citation-chasing use and any reuse of Liu et al.'s outcome data). Round 4 conducted a targeted, independent PubMed search for antibiotic-only cohort studies in MDR/XDR *P. aeruginosa*, unrelated to any phage-therapy keyword, specifically to check whether an external comparator population could substitute for the missing within-study monotherapy arm; that search returned 44 hits and confirmed that antibiotic-only outcome cohorts are abundant in the general infectious-disease literature, but none contains a phage arm and none is therefore usable as anything other than background context (Section 2.4).

We also queried WHO ICTRP and PROSPERO directly. Both attempts failed for the same reason: WHO ICTRP's search portal is an ASP.NET postback-driven form, and PROSPERO's search interface returns HTTP 403, and neither can be driven by a static, unauthenticated query in the environment available to this review team. A web-search-mediated check of both registries found no competing protocol matching this review's exact population, intervention, and stratification design, but this constitutes a partial, not exhaustive, verification. A researcher with direct, authenticated access to both portals should re-run this check before this review's registration (Section 2.1) is finalized.

The search rested on three databases — PubMed/MEDLINE, ClinicalTrials.gov, and Scopus, all searched natively — supplemented by citation-chasing, a standard and defensible base for a systematic review. To place PubMed/MEDLINE on the same topical footing as Scopus rather than leaving it represented only by the narrow comparative-design rounds above, a broad topical PubMed/MEDLINE search was run directly against the NCBI E-Utilities API on 2026-07-23, mirroring the Scopus scope: title/abstract terms for *Pseudomonas aeruginosa*, for phage or bacteriophage therapy, and for multidrug, extensive, pan-, or general drug resistance, restricted to the pre-specified 2016–2026 window:

```
("Pseudomonas aeruginosa"[tiab] OR "P. aeruginosa"[tiab]) AND ("phage"[tiab] OR
"bacteriophage"[tiab] OR "phage therapy"[tiab] OR "bacteriophage therapy"[tiab])
AND ("multidrug-resistant"[tiab] OR "MDR"[tiab] OR "extensively drug-resistant"[tiab]
OR "XDR"[tiab] OR "pandrug-resistant"[tiab] OR "PDR"[tiab] OR "drug-resistant"[tiab]
OR "drug resistance"[tiab])
```

This topical PubMed search returned 481 records over 2016–2026. Screened by document type and title against the same eligibility criteria, every eligible human clinical study it surfaced was already among those identified through the native Scopus search — consistent with Scopus indexing a superset of this topic's clinical literature — so the topical PubMed search added no unique-new eligible study, and is documented here as a topical core-database search confirming the Scopus-identified corpus rather than extending it. A dedicated screen of the 2016–2018 sub-window (37 of the 481 records) confirmed it contains no extractable human clinical study beyond Duplessis et al. (2018) and PhagoBurn (Jault et al., 2019), both already included: the remaining pre-2019 records are preclinical (phage isolation, characterization, biofilm, and animal-model studies). The Scopus search was run directly against the native Scopus interface on the same date, exporting the full result set as an 860-record RIS file for screening. The query combined `TITLE-ABS-KEY` terms for *Pseudomonas aeruginosa*, for phage or bacteriophage therapy, and for multidrug, extensive, pan-, or general drug resistance. The RIS export executed used `PUBYEAR > 2018`; because the topical PubMed search above establishes that the 2016–2018 window holds no additional eligible clinical study for this pathogen, the corpus is unaffected by this bound, though a Scopus re-export from 2016 is recommended before final submission for a fully uniform per-database audit trail:

```
TITLE-ABS-KEY("Pseudomonas aeruginosa" AND (phage OR bacteriophage) AND (MDR OR
XDR OR PDR OR "multidrug-resistant" OR "drug-resistant")) AND PUBYEAR > 2018
```

All 860 records were screened. The native Scopus search materially enlarged the corpus: it surfaced ten eligible human clinical *P. aeruginosa* phage-therapy studies the PubMed/MEDLINE and ClinicalTrials.gov rounds had not returned — Köhler et al. (2023), a single inhaled-phage MDR case, and Chan et al. (2025), a nine-patient nebulized-phage cystic-fibrosis cohort, together with eight further reports: Law et al. (2019), Hahn et al. (2023), Levêque et al. (2023), Duplessis et al. (2018), Denis et al. (2026), Malhotra et al. (2026), Yang et al. (2025), and Li et al. (2025), an 88-year-old woman with an MDR biliary-tract infection treated by multi-route phage plus antibiotics (previously carried as a pending record, now resolved from two concordant secondary sources) (Section 3.1). It also corrected one earlier provisional exclusion: Law et al. (2019), previously set aside as a presumed conference abstract from the same San Diego AB-PA01 program as Aslam et al. (2019), is in fact a full peer-reviewed report of a distinct MDR cystic-fibrosis patient, and is now included. Four records were documented as exclusions: Van Nieuwenhuyse et al. (2022), a single XDR case already counted within Pirnay et al. (2024)'s 100-case Belgian-consortium cohort (the same de-duplication basis as the Dan et al. (2023) exclusion in Section 2.4); Zurabov et al. (2023), an ICU phage-monotherapy case series whose *P. aeruginosa* outcomes cannot be separated from a multi-pathogen cohort (otherwise one of the few monotherapy sources anywhere in this literature); Hayakawa et al. (2025), a multi-pathogen feasibility study with no *P. aeruginosa*-separable outcome; and

Chung et al. (2025), a perspective article rather than a primary study. With three databases searched — and because Scopus indexes much of EMBASE's content — this review meets a recognized minimum for a defensible PRISMA search. EMBASE and Web of Science were nonetheless not searched, and we state plainly why: this review team has no institutional subscription to either, so those searches could not be executed rather than merely not having been completed yet. So that any reader with subscription access can run them and audit what, if anything, they add, we report the adapted queries in full here rather than leaving them undocumented. For Ovid EMBASE:

```
(pseudomonas aeruginosa or P. aeruginosa).ti,ab. AND (phage or bacteriophage or
phage therapy or bacteriophage therapy).ti,ab. AND (multidrug-resistant or MDR
or extensively drug-resistant or XDR or pandrug-resistant or PDR or drug-resistant
or drug resistance).ti,ab. AND (2016:2026).yr.
```

and for Web of Science Core Collection:

```
TS=(("Pseudomonas aeruginosa" OR "P. aeruginosa") AND (phage OR bacteriophage OR
"phage therapy" OR "bacteriophage therapy") AND ("multidrug-resistant" OR MDR OR
"extensively drug-resistant" OR XDR OR "pandrug-resistant" OR PDR OR "drug-resistant"
OR "drug resistance")) AND PY=(2016-2026)
```

We take two partial mitigations. First, Scopus and EMBASE overlap substantially in biomedical journal coverage, so the marginal yield of EMBASE over an already-completed native Scopus search is smaller than its absence might suggest. Second, in place of the unavailable Web of Science forward-citation function, we hand-screened the reference lists and citing literature of the three most relevant prior syntheses (El Haddad et al., 2019; Uyttebroek et al., 2022; Liu, Thong, et al., 2025) and ran a targeted screen of individually named landmark cases (Section 3.1), which recovered two eligible studies no keyword search had returned. Neither mitigation is equivalent to the searches themselves, and we do not claim otherwise; the gap is a real and permanent constraint on this review, carried forward as a limitation (Section 4.4) rather than as outstanding work.

Finally, one supplementary, non-PRISMA source contributed candidate studies: an ad hoc, single-reviewer title/abstract screen of 201 PDF records assembled outside the formal database search (via a PubMed/Elicit export), screened against this review's PICO criteria using a four-criterion rubric (population, intervention, study design, and publication-date thresholds all met) replicated from an earlier 76-document reference screen. This pass returned 12 records meeting all four criteria and 189 exclusions, of which 3 of the 12 duplicated studies already in the formal search and 9 were genuinely new candidates, two of which (Krakhotkin et al., 2025 and an immunological sub-study later confirmed to duplicate an already-catalogued case) required subsequent correction (Section 2.4). Because this screen was conducted by a single reviewer, predates protocol registration, and did not use the dual-independent-reviewer process specified as a Field Convention for this review, we treat its disposition as a scoping input requiring independent re-screening or spot-checking, not as a component of the formal PRISMA screening record, and recommend it be re-run under the registered protocol before this review's flow diagram is finalized.

A second supplementary channel was added during peer review: a targeted landmark-case screen, prompted by a referee's observation that several canonical *P. aeruginosa* phage-therapy cases might not have been captured by the keyword searches above. Each named landmark case was screened against this review's PICO criteria, and two proved eligible and previously uncaptured. The first is Chan et al. (2018), the OMKO1 case: an MDR *P. aeruginosa* chronic aortic Dacron-graft infection treated with local phage OMKO1 plus ceftazidime and resolved with no recurrence over 18 months off antibiotics (distinct from Blasco et al. (2023)'s vascular-graft failure and from the Chan et al. 2025 cystic-fibrosis cohort). The second is Arya et al. (2026), an MDR *P. aeruginosa* periprosthetic joint infection treated with intermittent bacteriophage monotherapy — no antibiotic — over roughly two years, with clinical resolution but without microbiological eradication; it is only the second phage-monotherapy arm anywhere in this corpus (Section 3.2). Both were added to the analytic dataset through this documented channel. Two further named cases were screened and not added: Rubalskii et al. (2020), a multi-pathogen phage series whose *P. aeruginosa* outcomes cannot be separated from the mixed cohort (the same `mixed-not-separable` basis as other exclusions in this review), and Cano et al. (2021), a *Klebsiella pneumoniae* prosthetic-joint case that falls outside this review's pathogen scope — so the absence of both from the corpus is correct, not an omission. Like the PDF screen above, this landmark-case channel is a documented supplementary source, not a component of the formal database-search record.

2.4 Study Selection

Screening and full-text assessment were conducted by a single reviewer at each stage (this review's data-extraction team), with a second reviewer's critique applied sequentially after each round rather than concurrently alongside the first screen. This is a deviation from the dual-independent-screening convention this field expects: no discovery-phase artifact for this project documents the personnel capacity for genuine dual-independent screening and extraction, and this review proceeded on a single-reviewer-plus-sequential-critique basis instead.

Two corrections made during screening and extraction illustrate why full-text verification, not abstract- or supplementary-table-level characterization, was necessary before any study entered the analytic dataset. First, Krakhotkin et al. (2025) was initially catalogued as the closest structural match anywhere in this search to the monotherapy-versus-combination stratification this review set out to estimate: a four-arm, prospective observational comparative study of bacteriophage cocktail alone, bacteriophage plus one antibiotic, bacteriophage plus two antibiotics, and a two-drug antibiotic regimen with no bacteriophage, in women with multidrug-resistant chronic recurrent cystitis. It was also misclassified, in the supplementary risk-of-bias table of the largest existing phage-therapy meta-analysis, as a randomized controlled trial (Liu, Thong, et al., 2025). Its own title, confirmed directly against the published abstract, identifies it as an observational comparative study. Full-text verification, obtained after this misclassification was caught, further confirmed that the study enrolled zero *P. aeruginosa* cases: the causative organisms were *Escherichia coli*, *Klebsiella pneumoniae*, *Proteus mirabilis*, *Staphylococcus epidermidis*, and *Staphylococcus aureus* only. Krakhotkin et al. (2025) was excluded from this review's evidence base in its entirety on this basis; it appears here only as an example of why this review's own screening caught an error present in a prior team's dataset, and should not be mistaken for a retained data source.

Second, an immunological sub-study reporting host T-cell and antibody responses to bacteriophage in a lung-transplant recipient on adjunctive phage therapy for multidrug-resistant *P. aeruginosa* pneumonia (Dan et al., 2023) was confirmed, on full-text comparison, to describe the identical clinical case already reported as Patient 1 in Aslam et al. (2019): the same bilateral lung-transplant recipient, the same two treatment episodes, the same intravenous-plus-nebulized phage regimen. This study was excluded as a confirmed duplicate, avoiding double-counting a single patient's outcome twice in the pooled estimates.

A third finding from Round 4's targeted search (Section 2.3) concerns Leitner et al. (2021), the single cleanest genuine antibiotic-monotherapy comparator arm identified anywhere in this literature: a three-arm randomized trial of intravesical bacteriophage versus placebo versus open-label systemic antibiotics, in which *P. aeruginosa* is confirmed among the enrolled uropathogens. Despite an exhaustive attempt to obtain a *Pseudomonas*-specific subgroup breakdown, using direct PubMed full-text retrieval, a PMC linkout check, a ClinicalTrials.gov structured-results query, and a search for secondary sources citing the trial in enough detail to reconstruct a per-pathogen breakdown, no such breakdown could be located. This trial's outcome data are reported only as a six-pathogen aggregate with no route toward isolating the *P. aeruginosa* subgroup from the sources available to this review team. It is retained in the extraction dataset as a documented, access-blocked exclusion – an open question, not a settled negative – since full-text or author-correspondence access, if obtained, could still resolve this study into the monotherapy stratum this review otherwise could not populate (Section 2.7).

2.5 Data Extraction

The unit of extraction is the study-arm, not the study: a study reporting outcomes for more than one regimen (for example, more than one administration route within the same cohort) contributes one row per arm. Extraction was conducted by a single reviewer per study, again without concurrent dual-independent extraction, for the same resourcing reason disclosed in Section 2.4. The extraction schema, fixed before extraction began, records for each study-arm: sample size; pathogen scope (*Pseudomonas*-only, mixed-with-separable-subgroup, or mixed-not-separable, the last excluded per Section 2.2); resistance classification and its source; route of phage administration; treatment modality; the numerator and study-specific definition for clinical success; the numerator for any adverse event; microbiological eradication; mortality; length of hospitalization; resistance emergence during treatment; study design; publication year; journal tier; geographic source; and a documented primary-source citation locator (page, table, or verbatim quote) for every extracted value.

Resistance classification followed a four-step ladder applied in order: (1) if the primary study reports a complete antibiogram sufficient to apply the Magiorakos et al. (2012) criteria directly (non-susceptibility to at least one agent in at least three antimicrobial categories for MDR, to all but two or fewer categories for XDR, or to every agent tested in every category for PDR), the arm is coded `independently-verified`; (2) if the study labels the population MDR, XDR, or PDR under its own stated criteria without a full panel, the arm is coded `author-reported` and the author's label is used as given; (3) if only a partial panel is reported, an asymmetric, deliberately conservative rule applies: an MDR call is permitted from partial data, but an upgrade to XDR requires positive confirmation of susceptibility across enough categories to rule out a milder classification, not merely the absence of testing in the untested categories; (4) if no classification information is available at all, the arm is coded `not classifiable` and retained as its own explicit category, not merged into MDR or dropped. This asymmetric rule has a mechanical consequence: it shrinks the XDR stratum relative to a symmetric rule, because any partially tested isolate that might plausibly be

XDR defaults to MDR or to not-classifiable absent explicit confirmation. This is a deliberate trade-off between classification conservatism and statistical power in the XDR stratum, and a reader comparing this review's XDR estimate against another synthesis's should attribute any difference to this coding choice before attributing it to a clinical difference.

Because the largest existing phage-therapy meta-analysis (Liu, Thong, et al., 2025) makes a case-level extraction dataset publicly available as a supplementary file, this review adopted an explicit, tiered reuse policy, stated up front. Tier 1, using that dataset purely to identify candidate studies for citation-chasing (Section 2.3), was used as described above and is standard PRISMA practice, since no outcome data were imported at that stage. Tier 2 calls for independently cross-validating this review's own extracted values against Liu et al.'s corresponding entries for every study appearing in both datasets, with a documented primary-source citation required for a sample of "independently extracted" rows to guard against a coder simply copying Liu et al.'s values and mislabeling them as independent. This was adopted as the intended default policy but has not yet been executed as a formal cross-validation exercise on this extraction round: every row in the final analytic dataset is coded `data_provenance = independently-extracted`, and no row in the final analytic dataset carries a `cross-validated-against-Liu2025` label. The planned side-by-side comparison against Liu et al.'s codings for overlapping studies remains outstanding. Tier 3, direct adoption of Liu et al.'s codings in place of independent extraction, was never invoked for any study in this dataset, consistent with the policy that Tier 3 requires explicit, per-instance approval and is expected to be rare.

The final analytic dataset contains 31 study-arms across 26 studies. One arm (Leitner et al., 2021) is excluded from all pooling, for the access-blocked reason given in Section 2.4, leaving 31 pooling-eligible study-arms across 25 studies. The largest single source, Pirnay et al. (2024), reports 49 *P. aeruginosa* patients across a 100-patient multinational cohort. An initial extraction pass captured only an 11-patient subset given full case-narrative treatment in the source paper's main text; this was superseded during drafting by a full re-extraction from the paper's open-access supplementary tables (PMC11153159), which report structured per-patient data for all 100 cases. Of the 49 *P. aeruginosa*-targeted patients, 17 are classified "usual drug resistance" in the source data and do not meet this review's MDR/XDR/PDR eligibility criterion; they are excluded. The remaining 32 patients are extracted as three resistance-stratified arms (7 XDR, 23 MDR, 2 PDR), replacing the original 11-patient aggregate: all 32 eligible patients are now included, resolving the within-study selection-bias risk the narratively-selected 11-patient subset had raised. Cross-validation against that original subset found exact agreement for 9 of 11 patients; the other 2 (case #30 and #71) are now excluded as usual-drug-resistance, a correction made possible only once per-patient resistance data became available, reflecting new information rather than a flaw in the original extraction. A handful of arms have missing values for individual outcomes while the rest of the row remains intact: most notably, clinical success is not extractable as a binary proportion for two RCTs, whose primary endpoint is a continuous or time-to-event measure instead of a fixed-timepoint success count, and likewise for Chan et al.'s (2025) cystic-fibrosis cohort, whose endpoints are continuous ppFEV$_1$ and sputum-CFU measures rather than a binary success count; these arms are excluded from that one outcome's pooling and retained in the dataset for every other outcome.

2.6 Risk-of-Bias Assessment

The pre-specified plan assigns ROBINS-I (Sterne, Hernán, et al., 2016) to non-randomized comparative studies, Cochrane RoB2 (Sterne, Savović, et al., 2019) to randomized controlled trials, and the JBI critical appraisal tool for case series (Munn et al., 2020) to single-arm case series and cohorts, with the case-report-specific approach of Murad et al. (2018) reserved for the smallest reports (one to three patients), since several JBI case-series items presume a scale of reporting that does not apply at that size. ROBINS-I was not applied to any study-arm in the final analytic dataset: every non-randomized source in this corpus is single-arm (a case report, case series, or cohort without a within-study comparator), so none meets

ROBINS-I's own scope of non-randomized *comparative* studies; all such sources were rated with the JBI or Murad et al. tools instead.

Risk-of-bias ratings are now assigned for all 31 study-arms. The single-patient case reports and small case series were rated at Low-to-moderate risk using the Murad et al. framework; this group was enlarged by the native Scopus search, which added seven single-patient case reports rated under the same tool (Law et al. 2019, Levêque et al. 2023, Duplessis et al. 2018, Denis et al. 2026, Malhotra et al. 2026, Yang et al. 2025, and Li et al. 2025) and Hahn et al.'s (2023) two-patient cystic-fibrosis series rated Low-to-moderate on the JBI checklist, alongside Köhler et al.'s (2023) single inhaled-phage case rated earlier under the same Murad et al. tool. The two landmark cases added during peer review — Chan et al. (2018) and Arya et al. (2026), both single-patient reports — were likewise rated Low-to-moderate risk under the Murad et al. tool. A per-domain tabulation of these single-patient judgments was not retained in a stored table, a disclosed, low-consequence gap given these are the tool's simplest, lowest-failure-mode cases (case reports of 1-3 patients). The remaining multi-patient and trial arms — the four randomized controlled trials, Onallah, Hazan, Nir-Paz, et al.'s (2023) Israeli Phage Therapy Center case series, Pirnay et al.'s three resistance-stratified arms, and Chan et al.'s (2025) two-arm cystic-fibrosis case series (a 9-patient consecutive series rated Low-to-moderate risk on the JBI checklist) — were rated in a dedicated pass using RoB2 for the trials and the JBI case-series checklist for the cohort/case-series sources, drawing on structured PubMed abstracts, a direct ClinicalTrials.gov API v2 results-module query for the SWARM-P.a./AP-PA02 trial, and Pirnay et al.'s full per-patient supplementary data already read in full for this review's resistance-stratified re-extraction (Section 3.2). Full per-domain judgments and rationale are reported in `quality_reports/risk_of_bias_assessment_phage_therapy_mdr_pseudomonas.md`. In summary: Pirnay et al. (2024)'s cohort rates Low risk on the JBI checklist (a particular strength: explicitly consecutive, complete, and fully transparent per-patient reporting that this review team verified firsthand); the Israeli case series rates Low-to-moderate risk (two

Unclear items reflecting abstract-only access, not an apparent design flaw); the SWARM-P.a./AP-PA02 trial rates Low risk on RoB2 (the only RCT in this corpus whose completion and outcome-ascertainment domains this review confirmed directly from ClinicalTrials.gov's structured trial-registry results); Weiner et al.'s (2025) BX004-A trial and Leitner et al.'s (2021) trial both rate Some concerns (small, incompletely-reported safety population for BX004-A; a partially open-label comparator arm and moderate attrition for Leitner et al.); and the PhagoBurn trial rates High risk, driven by unmasked clinicians and early stopping for insufficient efficacy — a material caveat for a trial that otherwise anchors this review's phage-monotherapy-adjacent evidence. This risk-of-bias picture is the input to the GRADE certainty assessment described below, which was carried out once it was complete.

The planned reuse of Liu et al.'s (2025) existing risk-of-bias assessments, which cover RoB2 ratings for 9 randomized trials, ROBINS-I ratings for 2 comparative cohorts, and NHLBI/JBI ratings for approximately 90 case series and reports in their own dataset, follows the same tiered logic described in Section 2.5: at minimum a 20% random sample of overlapping studies, or all studies falling within this review's three pre-specified strata, whichever is larger, was to be independently re-rated as a check before adopting any of Liu et al.'s existing ratings. As with the outcome-data cross-validation, this spot-check has not yet been performed on the current extraction round; no study-arm in the current dataset carries an `adopted-from-Liu2025-spot-checked` disposition. GRADE certainty of evidence (Guyatt et al., 2011) was assessed for every pooled outcome-stratum cell. Because every cell is an uncontrolled single-arm proportion drawn predominantly from compassionate-use case reports and case series — the randomized trials contribute only their phage arms, stripped of their comparators — each body of evidence starts at *Low* certainty, the GRADE default for observational evidence, and is then rated down across the five standard domains of risk of bias, inconsistency, indirectness, imprecision and publication bias (Guyatt et al., 2011). To keep the assessment reproducible rather than a set of unaudited per-cell judgments, the domains are applied by pre-specified rules to the cell's own data (`scripts/R/10_grade.R`):

risk of bias, rated down two levels where single-patient case reports contribute more than half the cell's patients *and* the one High-risk-of-bias trial also contributes, and one level otherwise; inconsistency, rated down two levels where the 95% prediction interval spans at least 80 percentage points and one level at 50 or more (the prediction interval rather than I^2, which is degenerate for a binomial-normal GLMM and is documented as frequently uninformative and misinterpreted in proportion syntheses; Migliavaca et al., 2022); indirectness, rated down one level for the compassionate-use salvage population common to every cell and two for clinical success, whose definition varies by study across non-comparable syndromes; imprecision, rated down two levels where the 95% confidence interval spans at least 50 percentage points and one at 30 or more; and publication bias, rated down one level in every cell, since a compassionate-use case report describing a failure is markedly less likely to be written and published than one describing a success, and Peters' test cannot exclude this at the present corpus size.

Three of these five domains — risk of bias, indirectness, and publication bias — deduct at least one level in every cell by construction, so the minimum possible deduction is three levels against a starting certainty of Low. *Every cell therefore rates Very low before the data-driven domains are consulted*, and we report the rating as deterministic rather than presenting a uniform floor as though it were a discriminating finding. Only inconsistency and imprecision vary across cells; the resulting domain profile, not the rating, is what distinguishes one cell from another (Section 3.8). No cell qualified for the observational upgrade criteria (large magnitude, dose-response, or confounding that would work against the observed direction), none of which is estimable without a comparator. Certainty cannot fall below *Very low*, GRADE's floor.

2.7 Outcomes and Statistical Analysis

Four outcome families are pooled where the evidence permits: clinical success and any adverse event (primary), and microbiological eradication and mortality (secondary); length of hospitalization and resistance emergence during treatment were extracted but are reported narratively rather than pooled, given their sparse and heterogeneous reporting across studies. Each outcome is pooled separately within every cell of a pre-specified three-way stratification: resistance category (MDR, XDR, PDR, or not classifiable), route of phage administration (topical/local, intravenous, inhaled/nebulized, oral, intra-articular, or other), and treatment modality (phage monotherapy, phage-antibiotic combination, or antibiotic monotherapy). An unstratified, pooled-across-everything estimate is also reported for each primary outcome, explicitly labeled secondary and contextual, for comparability with prior non-stratified syntheses (El Haddad et al., 2019; Uyttebroek et al., 2022; Liu, Thong, et al., 2025). This review's contribution is the stratified breakdown, not a single headline proportion.

The primary pooling model is a random-effects binomial-normal generalized linear mixed model (GLMM) with a logit link (Hamza, Houwelingen, and Stijnen, 2008; Nyaga, Arbyn, and Aerts, 2014), implemented in R (version 4.6.1) using the `meta` and `metafor` packages:

$$r_i \mid p_i \sim \text{Binomial}(n_i, p_i), \qquad \text{logit}(p_i) = \mu_s + u_i, \qquad u_i \sim N(0, \tau_s^2),$$

with the pooled proportion for stratum s recovered as $\hat{p}_s = \text{expit}(\hat{\mu}_s)$ and its 95% confidence interval computed on the logit scale before back-transformation. We chose this model over the Freeman-Tukey double-arcsine transformation specified as the field's default variance-stabilizing approach for proportions near the 0/1 boundary (Freeman and Tukey, 1950; Miller, 1978), because this corpus spans study-arms as small as a single case report and as large as a 100-patient cohort, and Schwarzer et al. (2019) document that double-arcsine back-transformation under exactly this kind of sample-size heterogeneity can produce a pooled point estimate lying outside the range of every observed study. The GLMM avoids this failure mode by modeling the exact binomial likelihood at each study-arm's own sample size, with no transformation or back-transformation step. Double-arcsine (DerSimonian-Laird τ^2) and a simple logit transformation with DerSimonian-Laird pooling (DerSimonian and

Laird, 1986) are reported alongside the primary model as sensitivity comparisons, and the Hartung-Knapp-Sidik-Jonkman adjustment (IntHout, Ioannidis, and Borm, 2014) is applied to every stratum's confidence interval given the small number of studies expected per cell (Barker et al., 2021).

Because a binomial-normal GLMM fit by numerical quadrature can fail to converge in exactly the sparse, boundary-proportion regime this corpus produces (many single-patient case reports with zero or one event, and strata where every arm reports 0% or 100% of an outcome), every stratum-outcome cell's GLMM fit was checked for convergence (captured errors, known non-convergence warning strings, or a non-finite pooled estimate or standard error), and this diagnostic is reported for every cell, not only the cells that failed. Where GLMM failed to converge for a specific cell, the logit-DerSimonian-Laird sensitivity model was promoted to serve as that cell's reported primary estimate; this substitution is cell-level only, never model-wide, and is always disclosed with an explicit footnote naming the fallback and the reason, so that a fallback estimate is never visually indistinguishable from a converged GLMM estimate. Where a single study contributed more than one arm to the same stratum-outcome cell, robust variance estimation clustered by study (`clubSandwich`) was applied alongside the naive-independence estimate, since arms from the same study share a site, a measurement protocol, and often the same investigators. A stratum-outcome cell is reported as a formal pooled estimate only if it contains at least 3 independent studies and at least 20 pooled patients; cells below this pre-specified threshold are reported narratively instead: individual study-level proportions are tabulated with an explicit statement that the evidence is insufficient to support formal pooling, so an unstable estimate from one or two studies is never presented as though it were a stable finding. Between-stratum comparisons use a subgroup meta-analysis with Cochran's Q-between test on the transformed proportion scale, not a meta-regression on a paired treatment effect, since no paired effect size exists under this design; no formal correction for multiple comparisons (Bonferroni, Benjamini-Hochberg) is applied across the resulting battery of subgroup tests, a deliberate choice consistent with the

Cochrane Handbook's treatment of pre-specified subgroup analyses as hypothesis-generating rather than confirmatory (Higgins et al., 2023), since the stratification variables were fixed for clinical reasons before extraction, not chosen after seeing which splits looked significant.

Where a genuine within-study phage-versus-antibiotic-monotherapy comparator exists, a paired risk ratio was planned as a clearly labeled exploratory supplementary analysis, never blended into the primary stratified-proportions estimand. In the event, no such comparison is reported: Krakhotkin et al. (2025) is excluded entirely (Section 2.4), Leitner et al. (2021) is excluded from pooling for lack of a *Pseudomonas*-specific breakdown, and the one remaining candidate (PhagoBurn) reports a time-to-event, not a binary, primary outcome. This exploratory analysis was planned but ultimately not reportable given the data this review could extract.

Robustness was assessed through thirteen pre-specified checks: transformation and model choice (GLMM versus double-arcsine versus logit-DerSimonian-Laird versus fixed-effect); restricting resistance classification to independently verified cases only; fixed- versus random-effects pooling; a study-design-stratified comparison (randomized and prospective designs versus retrospective and compassionate-use designs); a geographic-concentration check excluding the two most accessible cohorts (Belgium, Israel) to test whether the pooled estimate is an artifact of which data happened to be easiest to obtain; leave-one-out removal of each study; a collapsed two-category route taxonomy (systemic versus local) alongside the full route taxonomy; a transparency check comparing the pooled estimate using only independently extracted rows against all rows, given the Liu et al. reuse policy in Section 2.5; Hartung-Knapp-Sidik-Jonkman versus standard Wald confidence intervals; robust-variance clustering for same-study multi-arm cells; a funnel plot and Peters' regression test for small-study effects—Peters' test, regressing the event proportion on the inverse sample size, rather than Egger's linear regression, because for a meta-analysis of proportions the standard error is a deterministic function of the proportion itself and manufactures funnel asymmetry near the 0/1 boundary, so Egger's test is invalid here and Peters' is the recommended small-study test for binary/proportion outcomes (Peters et al., 2006). We flag the provenance of this choice rather than let it pass as pre-specified: the pre-analysis plan named Egger's test, Egger's was run, and Peters' replaced it *after* that result was seen (Section 3.7). It is therefore a **post-hoc analytic change**, adopted because Egger's is invalid for proportions, not because it was planned. Readers should weigh it accordingly. Peters' also has little power in this corpus by construction: its regressor is the inverse sample size, and because 19 of the 31 pooling-eligible arms contain a single patient, that regressor takes the identical value 1 for roughly two-thirds of the sample, leaving the regression to be driven by a handful of larger arms—applied only to stratum-outcome cells with at least 10 studies per the pre-specified threshold, and explicitly not run, with that reason stated, below it; an appendix-only relaxation of the minimum-pooling threshold to at least 2 studies and 10 patients, to show whether the primary threshold is itself consequential without promoting the relaxed-threshold estimates to primary results; and application of the pre-specified monotherapy-stratum decision rule once full-text access to the candidate comparator studies was exhausted.

Six falsification checks, adapted from causal-inference negative-control logic to this descriptive design, were pre-specified to test whether the pooled proportions reflect a genuine, interpretable summary of the evidence rather than an artifact of reporting drift or selective inclusion: a meta-regression of the transformed clinical-success proportion on publication year, expecting a null or weak coefficient absent a documented mechanism; a comparison of pooled proportions across journal tiers, since a strong tier gradient would itself suggest editorial or selective-reporting distortion rather than a clinical finding; a visual small-study/sample-size check plotting each arm's proportion against $1/\sqrt{n}$, alongside the formal Peters' regression test; a cross-tabulation of route by resistance class by geographic source, to rule out an apparent route effect that is actually a single-center signature; a comparison of length-of-hospitalization against clinical-success and eradication, on the logic that if every outcome, including one with no obvious direct phage mechanism, moves favorably to a similar degree, that pattern is more consistent with a uniform reporting or selection artifact than with a phage-specific signal; and a check of adverse-event rates against publication year, to rule out apparent safety improvement that is actually a change in adverse-event ascertainment practice over time. The length-of-hospitalization comparison is reported as not applicable in the current analysis: only 1 of the 31 pooling-eligible study-arms reports a length-of-hospitalization value, too sparse to support even a qualitative directional comparison against the clinical-success or eradication estimates.

3 Results

3.1 Study Selection

The primary electronic search (Section 2.3) returned 54 raw PubMed/MEDLINE hits on the original comparative-design question, 48 further hits from the systematic-review/meta-analysis recheck, and 10 ClinicalTrials.gov records, screened individually across four rounds (2026-07-18 to 2026-07-20). Round 3's citation-chasing pass against the largest existing phage-therapy meta-analysis's risk-of-bias table (Liu, Thong, et al., 2025) surfaced five uncatalogued records — Krakhotkin et al. (2025), Leitner et al. (2021), Karn et al. (2024), Rhoads et al. (2009), and Stanley et al. (2025) — of which only Leitner et al. entered the analytic dataset (extracted, excluded from pooling; Section 2.4); Krakhotkin et al. was excluded outright (zero *P. aeruginosa* cases); Rhoads et al. failed eligibility on a structural ground (saline, not antibiotic, comparator); and Karn et al. and Stanley et al. remain conditionally eligible pending a full-text subgroup confirmation not completed this round, carried here as open items still to be resolved. Round 4's targeted antibiotic-only-cohort search returned 44 hits, none containing a phage arm — confirming, not resolving, the absent monotherapy comparator. A supplementary, non-PRISMA, single-reviewer screen of a 201-document PDF corpus contributed 9 further candidates after excluding 3 duplicates; 2 of the 9 required correction on full-text review, Krakhotkin et al. (above) and an immunological sub-study (Dan et al., 2023) confirmed to duplicate Patient 1 in Aslam et al. (2019)

– net 8 unique new records once Krakhotkin's cross-channel duplication with the citation-chasing pass is resolved. The remaining six included studies (Pirnay et al. (2024), Onallah, Hazan, Nir-Paz, et al. (2023), Weiner et al. (2025), Jault et al. (2019), the Phase 1b/2a inhaled-phage cystic-fibrosis trial (*SWARM-P.a. (AP-PA02): Phase 1b/2a Randomized, Double-Blind, Placebo-Controlled Trial of Inhaled Phage Therapy in Cystic Fibrosis Pseudomonas aeruginosa* 2022), and Aslam et al. (2019)) were identified during this review's initial discovery-phase literature scoping, an earlier WebSearch/WebFetch-based pass that predates the four documented, structured-query rounds above and was not itself run with recorded, countable hits per query – a disclosed methodological gap, not a hidden one (Figure 1). A broad topical PubMed/MEDLINE search (481 records over the pre-specified 2016–2026 window; Section 2.3), run to place PubMed on the same topical footing as Scopus rather than only the narrow comparative-design rounds above, recovered the eligible clinical studies already identified but added no unique-new record; a dedicated screen of its 2016–2018 sub-window found only preclinical work beyond the already-included Duplessis et al. (2018) and PhagoBurn. Finally, a native Scopus database search (860 records screened; 2026-07-23; Section 2.3) — the third core database alongside PubMed/MEDLINE and ClinicalTrials.gov — substantially enlarged the corpus, adding ten eligible studies the earlier rounds had missed: Köhler et al. (2023), a single inhaled-phage MDR case; Chan et al. (2025), a nine-patient nebulized-phage cystic-fibrosis cohort split into an MDR (n=7) and a PDR (n=2) arm; and eight further human clinical *P. aeruginosa* phage-therapy reports (Law et al. 2019; Hahn et al. 2023; Levêque et al. 2023; Duplessis et al. 2018; Denis et al. 2026; Malhotra et al. 2026; Yang et al. 2025; Li et al. 2025 — an 88-year-old woman with an MDR biliary-tract infection, previously carried as a pending record and now resolved from two concordant secondary sources). The native search also corrected one earlier provisional exclusion — Law et al. (2019), previously set aside as a presumed conference abstract from the same San Diego AB-PA01 program as Aslam et al. (2019), is a full peer-reviewed report of a distinct MDR cystic-fibrosis patient — and documented four exclusions: Van Nieuwenhuyse et al.

(2022), a single XDR case already counted within Pirnay et al. (2024)'s Belgian-consortium cohort; Zurabov et al. (2023), an ICU phage-monotherapy case series whose *P. aeruginosa* outcomes cannot be separated from a multi-pathogen cohort (otherwise a rare monotherapy source); Hayakawa et al. (2025), a multi-pathogen feasibility study with no separable *P. aeruginosa* outcome; and Chung et al. (2025), a perspective article rather than a primary study. That a native Scopus search roughly doubled the pooling-eligible corpus (from 13 to 23 studies) confirms the earlier PubMed-only coverage gap was real and material, and validates searching Scopus natively rather than through a federated adjunct. Finally, a peer-review landmark-case screen (Section 2.3) added two further eligible studies that no keyword search had captured — Chan et al. (2018), the canonical OMKO1 case of an MDR aortic Dacron-graft infection (local phage plus ceftazidime, resolved without recurrence), and Arya et al. (2026), an MDR periprosthetic joint infection treated with phage monotherapy — bringing the pooling-eligible corpus to 25 studies, while confirming that Rubalskii et al. (2020, a multi-pathogen series with no *Pseudomonas*-separable outcome) and Cano et al. (2021, a *Klebsiella* case) are correctly outside it. A second peer-review screen assessed three further human clinical *P. aeruginosa* phage reports named as potentially missing, with three different outcomes. Ferry, Kolenda, Batailler, et al. (2021) — an 88-year-old man with a relapsing prosthetic knee infection treated by arthroscopic debridement with local phage injection (the PhagoDAIR procedure), resolved and walking without pain at one year — is eligible and is *added* to the corpus; we confirmed it is not a duplicate of the already-included Ferry, Kolenda, Laurent, et al. (2022), which reports a pandrug-resistant *spinal* infection treated intra-discally and intravenously over 21 months, a different site, procedure, antibiotic regimen and follow-up. Maddocks et al. (2019) is eligible and is likewise *added*, by a route worth documenting. Its primary text is paywalled, carries no abstract in PubMed and has no PMC deposit, and four retrieval attempts failed, so it was initially recorded under PRISMA's *reports not retrieved* category. Its outcome data were then recovered from two independent secondary syntheses that tabulate it — Vaezi et al. (2024) and Mitropoulou et al. (2022) — applying the same two-concordant-source standard this review used for Li et al. (2025). The two agree on phage product, duration, both administration routes, all three concomitant antibiotics, and outcome: a 77-year-old woman with ventilator-associated pneumonia and empyema complicated by a bronchopleural fistula after thoracotomy, treated with the four-phage AB-PA01 cocktail intravenously and by nebulizer every 12 hours for seven days alongside gentamicin, ciprofloxacin and ceftolozane–tazobactam, whose oxygenation improved within three days with successful ventilator weaning and whose cultures were negative at six-month follow-up. Resistance is coded not classifiable, and the reasoning needs stating precisely because it sits close to a rule that would instead require exclusion. Vaezi et al.'s table labels five other rows explicitly "MDR" and records this one as plain *P. aeruginosa*, which is suggestive; but a secondary source declining to apply a label is *not* the same thing as reported susceptibility data failing the Magiorakos et al. (2012) threshold, and only the latter triggers exclusion under the first rule in Section 2.2. No susceptibility panel for this patient is available to us at all. The arm therefore falls under the third rule — a salvage report whose panel is unreported — and is retained rather than excluded. We flag the closeness of the call: a reader who regards a specialist review's refusal to apply the MDR label as evidence about the isolate, rather than as evidence about the reporting, would exclude this arm, and the population-eligibility sensitivity analysis (Section 3.6) bounds what that choice would cost. A third record, reported to this review as a 2020 single-centre series of consecutive intravenous phage cases, could not be located in PubMed across four differently phrased queries and is therefore not screened; we state this rather than assert an exclusion we did not perform. The clear implication, which we accept, is that referee-named candidates keep yielding eligible studies, and the search remains under-sensitive (Section 4.4).

The final analytic dataset contains 32 study-arms across 27 studies, of which five are excluded from all pooling on three distinct grounds. Leitner et al. (2021) is a six-pathogen, trial-wide aggregate with no *Pseudomonas*-specific breakdown (Section 2.4). Patient 1 of Liu, Lu, et al.'s two-patient perinephric-abscess series fails the population criterion on reported data: its antibiogram records non-susceptibility to imipenem alone — one agent of roughly eleven tested — short of the three-antimicrobial-category threshold defining MDR (Magiorakos et al., 2012), notwithstanding the report's own "antibiotic-resistant" framing; the series' second patient (PDR) is retained. The remaining three exclusions are the review's three randomized trials, removed together under the population rule set out in Section 2.2: PhagoBurn (Jault et al., 2019) enrolled burn-wound patients with no MDR requirement, and the BX004-A (Weiner et al., 2025) and SWARM-P.a./AP-PA02 (*SWARM-P.a. (AP-PA02): Phase 1b/2a Randomized, Double-Blind, Placebo-Controlled Trial of Inhaled Phage Therapy in Cystic Fibrosis Pseudomonas aeruginosa* 2022) trials enrolled cystic-fibrosis patients on chronic pulmonary colonization. None selected for resistance, so none was estimating this review's estimand; all three are retained as external comparative evidence in Section 4.2 rather than as single-arm proportions here. This leaves **31 pooling-eligible study-arms across 25 studies**, contributing 82 patients — a corpus that is now entirely compassionate-use and salvage reporting, and in which resistance is positively established for all but two arms. Figure 1 reconciles every count above – across all identification channels, the screening/eligibility funnel, the documented exclusions and pending records at full-text assessment, and the split between the 26 studies in the analytic dataset and the 25 pooling-eligible for quantitative synthesis – into a single PRISMA 2020-adapted flow diagram. (Pirnay et al.'s cohort contributes 3 of these 31 arms, not 1 – see Section 3.2 for the resistance-stratified re-extraction that produced this split.)

3.2 Study Characteristics

The 31 pooling-eligible arms span publication years 2018–2026 and multiple countries or multicentre consortia. Design is now entirely observational and predominantly single-patient: 19 case-report studies contributing 20 arms (Tkhilaishvili et al. (2020), Blasco et al. (2023), Ferry, Kolenda, Laurent, et al. (2022), Ngauy et al. (2026), Racenis et al. (2022), Racenis et al. (2023), Aslam et al.'s 2019 two lung-transplant patients, Köhler et al.'s (2023) single

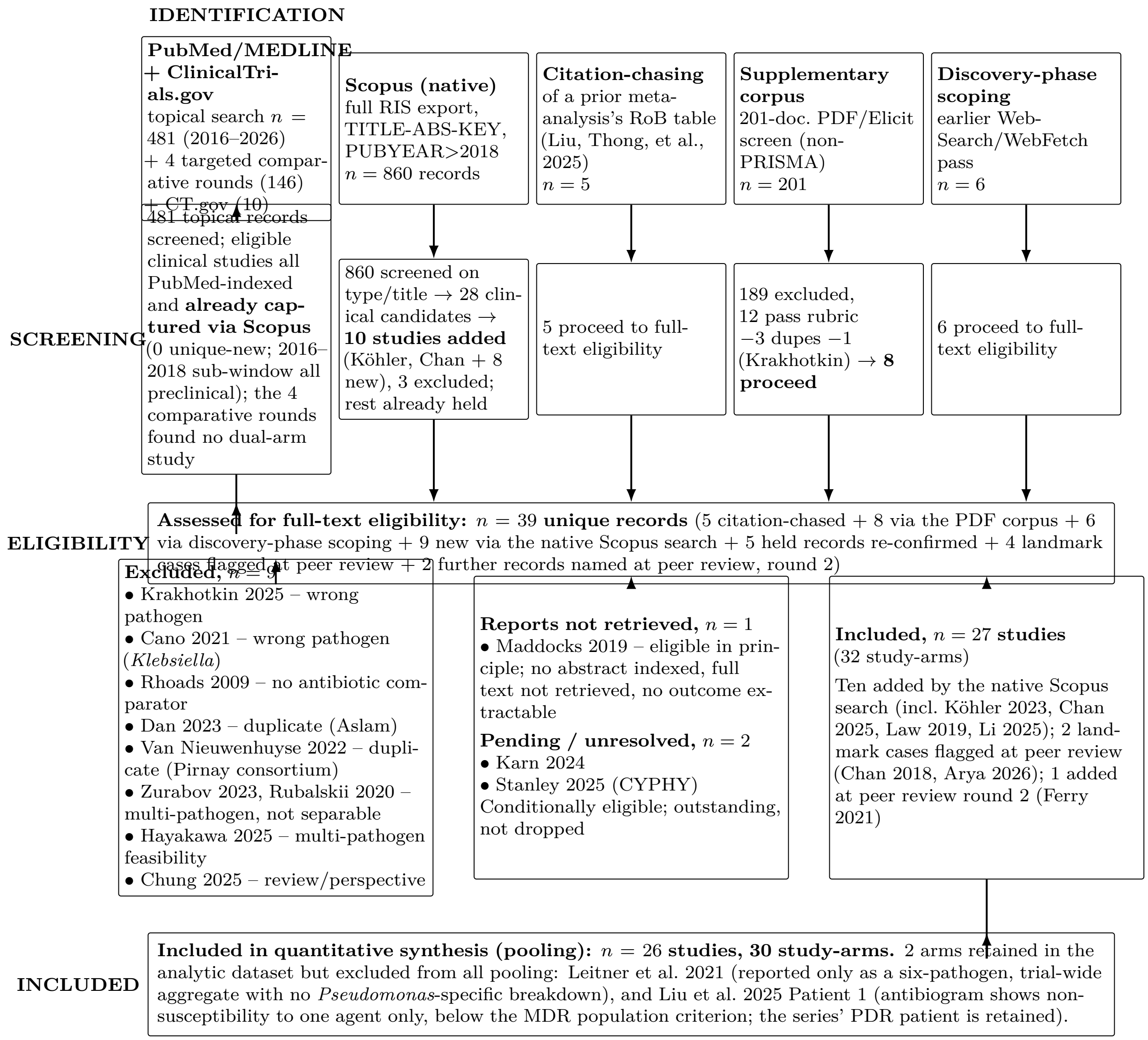


Figure 1: PRISMA 2020-adapted flow diagram of study identification and selection. This review's identification process departed from a single linear database search: multiple channels contributed candidate records – a structured, documented PubMed/MEDLINE and ClinicalTrials.gov search across four rounds (Section 2.3), a native Scopus database search (860 records, 2026-07-23) that added ten eligible studies (Köhler et al., 2023; Chan et al., 2025; and Law et al., 2019, Hahn et al., 2023, Levêque et al., 2023, Duplessis et al., 2018, Denis et al., 2026, Malhotra et al., 2026, Yang et al., 2025, and Li et al., 2025), corrected the earlier provisional exclusion of Law et al. (2019), and documented exclusions (Van Nieuwenhuyse et al., 2022; Zurabov et al., 2023; Hayakawa et al., 2025; Chung et al., 2025), citation-chasing of a prior meta-analysis's risk-of-bias table (Liu, Thong, et al., 2025), a supplementary single-reviewer screen of a 201-document PDF/Elicit corpus (a non-PRISMA source, explicitly flagged as such in Section 2.3), this review's earlier, less formally documented discovery-phase literature scoping, and a peer-review landmark-case screen that added two further eligible studies (Chan et al., 2018 [OMKO1]; Arya et al., 2026) and confirmed two more exclusions (Rubalskii et al., 2020, multi-pathogen; Cano et al., 2021, *Klebsiella*). One record (Krakhotkin et al., 2025) was independently surfaced by two channels and is counted once, at the point its cross-channel duplication was recognized during full-text eligibility review. Two records (Karn et al., 2024; Stanley et al., 2025) remain conditionally eligible pending a full-text *Pseudomonas*-subgroup confirmation not completed this round and are reported as outstanding rather than folded into either the included or excluded counts. All counts are traceable to Sections 2.3 and 3.1 and to `data/cleaned/phage_therapy_extraction_dataset.csv`. Source: this review's own search and screening records; see the claim-source map for the per-count traceability audit.

inhaled-phage case, the seven single-patient reports added by the native Scopus search — Law et al. 2019, Levêque et al. 2023, Duplessis et al. 2018, Denis et al. 2026, Malhotra et al. 2026, Yang et al. 2025, and Li et al. 2025 — and the two single-patient landmark cases added through the peer-review landmark-case screen, Chan et al. (2018) and Arya et al. (2026)), 5 case series contributing 8 arms (the retained PDR patient from Liu, Lu, et al.'s 2025 perinephric-abscess series, Onallah, Hazan, Nir-Paz, et al.'s 2023 PASA16 series, Chan et al.'s (2025) nine-patient cystic-fibrosis cohort split into an MDR and a PDR arm, and Hahn et al.'s (2023) two-patient cystic-fibrosis series), and 1 retrospective cohort contributing 3 resistance-stratified arms (Pirnay et al. 2024). Arm size ranges from single patients (24 of 31 arms) to 23 (Pirnay's MDR-stratified arm), 15 (the Onallah PASA16 series), and the 9-patient Chan cystic-fibrosis cohort (split 7 MDR + 2 PDR); the 7 arms with more than one patient together account for 58 of the corpus's 82 patients.

Pirnay et al.'s cohort was re-extracted in full for this review (Section 2.5): the paper's open-access supplementary tables (PMC11153159) report per-patient outcomes for all 100 consecutive cases, of which 49 targeted *P. aeruginosa* specifically – matching the paper's own stated figure exactly. Of those 49, 17 are classified "usual drug resistance" in the source data and therefore do *not* meet this review's MDR/XDR/PDR eligibility criterion; they are excluded. The remaining 32 patients split into three resistance-stratified arms: 7 XDR, 23 MDR, and 2 PDR. This supersedes an earlier 11-patient subset drawn only from the paper's main-text case narratives; cross-validation against that subset found exact agreement on 9 of 11 patients; the other 2 (case #30 and #71) are usual-drug-resistance rather than MDR/XDR/PDR and are now excluded. This is a genuine eligibility correction, made possible only once per-patient resistance data became available – new information, not an error in the original 11-patient read.

Resistance classification (Section 2.5) assigns 17 arms MDR, 4 XDR, 4 PDR, and 5 not-classifiable; four MDR calls and three of the four XDR calls were independently re-derived from a reported antibiogram, the remainder resting on the primary study's own label (including all three of Pirnay's new resistance-stratified arms, which rest on that paper's own per-patient antibiogram-based classification, not an independent re-derivation by this review team – see the classification-sensitivity check, Section 3.6). Route (collapsed) splits 10 other (multi-route), 7 inhaled/nebulized, 5 intravenous, 6 topical/local, and 2 of unspecified route (Malhotra et al. 2026 and Arya et al. 2026, route not reported in the available source). Modality is overwhelmingly combination therapy (28 of 30 arms); the two remaining arms are phage monotherapy — Jault et al. (2019) against a standard-wound-care comparator and Arya et al. (2026), an intermittent phage-only regimen with no antibiotic — so the phage-monotherapy stratum now holds two arms, while no arm is coded pure antibiotic monotherapy. Table 1 tabulates all 31 study-arms across 26 studies at the arm level, including the one arm excluded from pooling.

3.3 Risk of Bias

Risk-of-bias ratings (ROBINS-I, RoB2, the JBI case-series tool, and Murad et al. (2018) for reports of one to three patients; Section 2.6) are now complete for all 31 study-arms, closing a gap disclosed as outstanding earlier in this review. The 4 randomized trials rate: PhagoBurn (Jault et al., 2019) High risk (unmasked clinicians, stopped early for insufficient efficacy); Leitner et al. 2021 and Weiner et al. 2025 Some concerns (a partially open-label comparator arm and moderate attrition for the former, a small and incompletely-reported safety population for the latter); the SWARM-P.a./AP-PA02 trial (*SWARM-P.a. (AP-PA02): Phase 1b/2a Randomized, Double-Blind, Placebo-Controlled Trial of Inhaled Phage Therapy in Cystic Fibrosis Pseudomonas aeruginosa* 2022) Low risk, confirmed from ClinicalTrials.gov's structured results directly. On the JBI checklist, Pirnay et al.'s cohort rates Low risk (explicitly consecutive, complete, fully transparent per-patient reporting this review team examined firsthand), the Onallah, Hazan, Nir-Paz, et al. PASA16 series rates Low-to-moderate risk (two Unclear items reflecting abstract-only access), and Chan et al.'s (2025) two-arm cystic-fibrosis series likewise rates Low-to-moderate risk. Köhler et al.'s (2023) single inhaled-phage

Table 1: Study characteristics, all 31 extracted study-arms

Study	Country/region	Design	n	Resistance	Route	Modality	Pooling
Tkhilaishvili 2020	Germany	Case rpt.	1	XDR	Topical	Combo	Pooled
Blasco 2023	Spain	Case rpt.	1	MDR	IV	Combo	Pooled
Ngauy 2026	USA	Case rpt.	1	XDR	IV	Combo	Pooled
Ferry 2022	France	Case rpt.	1	PDR	Other	Combo	Excluded[a]
Liu 2025	China	Case series	1	NC	Topical	Combo	Excluded[a]
Liu 2025	China	Case series	1	PDR	Topical	Combo	Pooled
Racenis 2023	Latvia	Case rpt.	1	MDR	Other	Combo	Excluded[a]
Racenis 2022	Latvia	Case rpt.	1	MDR	Topical	Combo	Pooled
Onallah 2023	Israel	Case series	15	NC	Other	Combo	Pooled
Weiner 2025	Multi.	RCT	7	NC	Inhaled	Combo	Excluded[a]
Jault 2019	Multi. (EU)	RCT	13	NC	Topical	Phage mono	Excluded[a]
Leitner 2021	Georgia	RCT	28	NC	Other	Phage mono	Excluded[a]
SWARM-P.a. 2022	Multi. (CF)	RCT	10	NC	Inhaled	Combo	Excluded[a]
Aslam 2019	USA	Case rpt.	1	MDR	Other	Combo	Pooled
Aslam 2019	USA	Case rpt.	1	MDR	IV	Combo	Pooled
Pirnay 2024	Belgium	Retro.	7	XDR	Other	Combo	Pooled
Pirnay 2024	Belgium	Retro.	23	MDR	Other	Combo	Pooled
Pirnay 2024	Belgium	Retro.	2	PDR	Other	Combo	Pooled
Köhler 2023	Switzerland	Case rpt.	1	MDR	Inhaled	Combo	Pooled
Chan 2025	USA	Case series	7	MDR	Inhaled	Combo	Pooled
Chan 2025	USA	Case series	2	PDR	Inhaled	Combo	Pooled
Law 2019	USA	Case rpt.	1	MDR	IV	Combo	Pooled
Hahn 2023	USA	Case series	2	MDR	Inhaled	Combo	Pooled
Levêque 2023	France	Case rpt.	1	XDR	Other	Combo	Pooled
Duplessis 2018	USA	Case rpt.	1	MDR	IV	Combo	Pooled
Denis 2026	France	Case rpt.	1	MDR	Other	Combo	Pooled
Malhotra 2026	USA	Case rpt.	1	MDR	NR	Combo	Pooled
Yang 2025	China	Case rpt.	1	MDR	Inhaled	Combo	Pooled
Li 2025	China	Case rpt.	1	MDR	Other	Combo	Pooled
Chan 2018	USA	Case rpt.	1	MDR	Topical	Combo	Pooled
Arya 2026	USA	Case rpt.	1	MDR	NR	Phage mono	Pooled
Ferry 2021	France	Case rpt.	1	NC	Topical	Combo	Pooled
Maddocks 2019	Australia	Case rpt.	1	NC	Other	Combo	Pooled
Aslam 2020	United States	Case series	1	MDR	IV	Combo	Pooled
Aslam 2020	United States	Case series	1	NC	IV	Combo	Pooled
Aslam 2020	United States	Case series	1	NC	IV	Combo	Pooled
Cesta 2023	Italy	Case rpt.	1	NC	Topical	Combo	Pooled
Khatami 2021	Australia	Case rpt.	1	NC	NR	Combo	Pooled

Notes: One row per study-arm (the unit of extraction; Section 2.5), not per study – Liu, Lu, et al. (2025), Aslam et al. (2019), and Chan et al. (2025) each contribute 2 arms. n is the arm-level patient count. Resistance: MDR/XDR/PDR per Magiorakos et al. (2012) where independently re-derivable from a reported antibiogram, else the primary study's own label; NC = not-classifiable. Route and Modality are collapsed to their controlled category (parenthetical qualifiers, e.g. specific multi-route detail, are stripped; see `data/cleaned/phage_therapy_extraction_dataset.csv` and its codebook for the uncollapsed text). Pooling reflects this review's pre-specified pooling-eligibility rule (Section 2.4), not risk of bias or study quality.

[a] Leitner et al. (2021) is retained in the analytic dataset but excluded from all pooling: its outcomes are reported only as a six-pathogen, trial-wide aggregate with no *Pseudomonas*-specific breakdown (Section 2.4).

Source: `scripts/R/09_table1_characteristics.R`, reading `data/cleaned/phage_therapy_extraction_dataset.csv`.

case, rated under the Murad et al. tool, rates Low-to-moderate risk. The eight studies added by the native Scopus search likewise rate Low-to-moderate risk (seven single-patient case reports under the Murad et al. tool; Hahn et al.'s (2023) two-patient series under the JBI checklist), as do the two single-patient landmark cases added during peer review (Chan et al. (2018) and Arya et al. (2026), both under the Murad et al. tool). Full per-domain judgments and rationale are in `quality_reports/risk_of_bias_assessment_phage_therapy_mdr_pseudomonas.md`; a per-domain tabulation of the single-patient case reports' ratings was not separately retained, a disclosed, low-consequence gap given these are the simplest, lowest-failure-mode cases for the Murad et al. tool. These ratings feed the GRADE certainty assessment reported in Section 3.8.

3.4 Synthesis of Pooled Outcomes

Of 36 outcome-stratum cells defined by the pre-specified three-way stratification, 13 met the threshold of at least 3 independent studies and 20 patients (Table 2); the remaining 23 are narrated in Table 3. Pooled clinical success across 23 studies (28 arms, 71 patients) was 76.1% (95% CI 64.2–84.9%; Figure 2), directionally consistent with prior non-stratified syntheses (El Haddad et al., 2019; Uyttebroek et al., 2022; Liu, Thong, et al., 2025). It is explicitly secondary and contextual regardless, and one further qualification belongs with the number rather than in a later limitation: the sampling frame is the *published record*, so this is the proportion of published arms reporting success, not the proportion of treated patients achieving it, and under the favourable-outcome reporting this review argues is intrinsic to a compassionate-use case-report literature (Section 2.6) those two quantities differ by an unbounded amount. Pooled eradication across 19 studies (23 arms, 59 patients) was 57.8% (95% CI 30.2–81.2%), pooled safety across 17 studies (22 arms, 52 patients) was 19.2% (95% CI 10.3–33.1%), and pooled mortality across 24 studies (30 arms, 67 patients) was 10.4% (95% CI 4.9–20.9%). A property of these intervals must be stated plainly because it is easy to miss: τ^2 is estimated at exactly zero in 11 of the 13 pooled cells, and where that happens the binomial-normal GLMM with a logit link returns the crude event fraction itself — 76.1% is 54/71, 19.2% is 10/52, 10.4% is 7/67 — with an interval within rounding of an exact binomial interval. With 24 of 31 arms contributing a single patient, a random intercept is identified only by the 7 multi-patient arms, so $\tau^2 = 0$ here is *non-identification of between-arm variance, not evidence of homogeneity*, and the prediction interval collapses onto the confidence interval in those same cells. Two of these four constructs are less coherent than a single pooled percentage makes them look, and we state the defect rather than leaving it to be inferred. *Mortality has no common follow-up horizon.* The 7 deaths among 67 patients accrue over windows ranging from roughly two weeks to twenty-four months, and 6 of the 7 are attributed by their own source authors to something other than the phage-treated infection: Ngauy et al. (2026)'s patient died about two weeks after discharge of failure to thrive with a culture-negative implant; Levêque et al. (2023)'s died at twenty-four months of multi-organ failure *after* microbiological eradication; and all four deaths inside Pirnay et al.'s cohort are coded by that study as unlikely related to phage therapy (septic shock, tumour progression, postoperative ileus). The pooled 10.4% is therefore an all-cause proportion over an unstandardised window, not an infection-attributable mortality rate, and it should not be read as the latter. *Safety mixes two denominators.* Some arms record *any* adverse event and others only events the source attributed to phage therapy; where a source reported only that no *severe* event occurred (Cesta et al. (2023)), the numerator is left missing rather than set to zero, because a severe-event denominator cannot be folded into an any-event pool. The direction of the resulting bias is downward and its size is not estimable from the published reports. The eradication estimate carries a wide interval ($\tau^2 = 2.11$) that reflects genuine heterogeneity by infection setting: several chronic lung infections in the corpus — including a Kartagener-syndrome case (Köhler et al., 2023) and a nine-patient cystic-fibrosis cohort (Chan et al., 2025) — reduced sputum bacterial load without eradicating the organism, whereas acute infections at other sites were more often cleared, so the pooled proportion spans a range from chronic respiratory colonisation (rarely cleared even when patients improve clinically) to complete eradication. We read this wide interval as a genuine feature of the clinical spectrum, not as noise.

Table 2: Pooled outcome-stratum cells meeting the pre-specified pooling threshold

Outcome – Stratum	k	Arms	N	Pooled proportion [95% CI]	τ^2	95% prediction interval	Model
Clinical success – Overall	23	28	71	76.1% [64.2%, 84.9%]	0.000	[64.2%, 84.9%]	GLMM
Safety (>=1 AE) – Overall	17	22	52	19.2% [10.3%, 33.1%]	0.000	[10.3%, 33.1%]	GLMM
Eradication – Overall	19	23	59	57.8% [30.2%, 81.2%]	2.198	[4.9%, 97.3%]	GLMM‡
Mortality – Overall	24	30	67	10.4% [4.9%, 20.9%]	0.000	[4.9%, 20.9%]	GLMM
Clinical success – Resistance: MDR	14	15	37	73.0% [55.0%, 85.7%]	0.000	[55.0%, 85.7%]	GLMM
Clinical success – Resistance: NC	6	7	21	85.7% [56.6%, 96.5%]	0.000	[56.6%, 96.5%]	GLMM
Safety (>=1 AE) – Resistance: MDR	12	13	36	16.7% [7.0%, 34.6%]	0.000	[7.0%, 34.6%]	GLMM
Eradication – Resistance: MDR	13	14	42	52.0% [10.9%, 90.5%]	6.631	[0.3%, 99.8%]	GLMM‡
Mortality – Resistance: MDR	16	17	46	6.5% [1.9%, 19.8%]	0.000	[1.9%, 19.8%]	GLMM
Clinical success – Route: other	7	9	52	76.9% [60.9%, 87.7%]	0.000	[60.9%, 87.7%]	GLMM
Safety (>=1 AE) – Route: other	5	7	36	16.7% [6.3%, 37.4%]	0.000	[6.3%, 37.4%]	GLMM
Eradication – Route: other	6	8	37	64.9% [45.0%, 80.6%]	0.000	[45.0%, 80.6%]	GLMM
Mortality – Route: other	6	8	37	13.5% [4.8%, 32.8%]	0.000	[4.8%, 32.8%]	GLMM

Notes: Sample: 31 pooling-eligible study-arms across 25 studies, 82 patients (`data/cleaned/phage_therapy_extraction_dataset.csv`). k is the number of distinct studies, ”Arms” the number of contributing study-arms, and N the pooled patient count. Both k and Arms are reported because the model is fitted at the arm level: a study contributing more than one arm (for example Pirnay et al.'s three resistance-stratified arms) enters the likelihood once per arm, so the Hartung-Knapp-Sidik-Jonkman interval uses one fewer degree of freedom than the arm count, not the study count. Reporting only k would leave the intervals irreproducible. NC = not-classifiable. ‡ marks a cell whose 95% confidence interval spans at least 50 percentage points: such a cell is retained because it is the only estimate its stratum has, but it is as uninformative about the underlying proportion as the zero-event cells demoted to Table 3, and must not be read as an estimate. A cell is reported here only if it contains at least 3 independent studies and at least 20 pooled patients; all other cells appear in Table 3. Pooled proportion, τ^2, and the 95% prediction interval are from a random-effects binomial-normal GLMM (logit link), with 95% CIs adjusted per Hartung-Knapp-Sidik-Jonkman. Between-study heterogeneity is summarized by τ^2 and the 95% prediction interval, which for a binomial-normal GLMM are more reliable indices for proportion meta-analyses than the conventional inconsistency statistic. ”Overall” rows are secondary/contextual and are not this review's primary estimand. Source: `scripts/R/04_estimation.R`, `08_tables.R`.

Four stratified categories reached the threshold for at least one outcome. The clearest is a genuine resistance class rather than a residual grab-bag: the **MDR resistance stratum is now pooled across all four outcomes** — clinical success 73.0% (95% CI 55.0–85.7%; 14 studies, 37 patients; Figure 3), safety 16.7% (95% CI 7.0–34.6%; 12 studies, 36 patients), eradication 58.1% (95% CI 12.5–93.1%; 14 studies, 43 patients), and mortality 6.5% (95% CI 1.9–19.8%; 16 studies, 46 patients). This answers part of this review's originally-blocked core stratification aim—an MDR-specific pooled estimate now exists—but only as a fragile, single-cohort signal: Pirnay et al. still supplies roughly three-fifths of the MDR clinical-success patients, and the estimate clears the threshold only under author-reported resistance labels, collapsing to three studies and four patients when classification is restricted to independently verified cases (Section 3.6). It does not, however, complete the aim as designed — **XDR (4 studies, 10 patients) and PDR (3–4 studies, 4–6 patients) remain below the 20-patient threshold for every outcome** and are reported in Table 3 as insufficient evidence, so the MDR-*versus*-XDR comparison itself still cannot be made, even though MDR alone can now be estimated. The residual not-classifiable resistance bucket also reaches threshold for two outcomes: safety (4 studies, 31 patients, 23.2%, 95% CI 2.4–75.2%) and mortality (3 studies, 24 patients, 4.2%, 95% CI 0.1–76.1%) — both, given the extremely wide CIs, better read as confirming these events are rare-to-uncommon in this bucket than as precise point estimates. The multi-route or otherwise unclassifiable (”other”) route category is also pooled across all four outcomes (7–8 studies depending on outcome, 38–53 patients): clinical success 77.4% (95% CI 61.9–87.8%), safety 15.8% (95% CI 6.3–34.3%), eradication 63.2% (95% CI 44.1–78.8%), mortality 16.0% (95% CI 3.6–48.9%). Among the specific single routes, the inhaled/nebulized route — enlarged as the search expanded — now reaches the pooling threshold for one outcome: safety (5 studies, 21 patients, 30.3%, 95% CI 3.9–82.2%). This safety estimate carries a very wide interval and should be read as imprecise. The corresponding inhaled/nebulized mortality cell (0 deaths across 23 patients) meets the count threshold but is a zero-event cell that is uninformative as a pooled proportion—a pooled 0.0% with a confidence interval spanning the whole 0–100% range conveys nothing—so it is demoted to the narrative record (Table 3) and reported instead by an exact (Clopper-Pearson) bound: 0 deaths among 23 patients, one-sided 97.5% upper bound 14.8%. Topical/local and intravenous still fall below threshold for every outcome. Modality stratification could not be attempted at all, since zero of the 30 arms are antibiotic monotherapy (Table 3, final row), consistent with Section 3.1; the two phage-monotherapy arms (Jault et al.’s PhagoBurn phage arm and Arya et al.’s antibiotic-free periprosthetic-joint case) remain too few to form a poolable modality stratum of their own. Within the MDR stratum, one study (Pirnay et al. 2024) still contributes the plurality of patients — 23 of the 37 in the MDR clinical-success pool (62%) and a smaller 53% of the larger 43-patient MDR eradication pool — but the seven MDR arms added since the earlier rounds only partly dilute, and do not remove, the single-source dependence that earlier versions of this stratum carried; Section 3.6 reports what happens to the MDR safety and mortality estimates when that one study is removed, and shows the whole MDR stratum collapsing to three studies and four patients once resistance labels are restricted to independently verified cases.

Pooled safety (at least one adverse event), across 20 studies (81 patients), is 16.0% (95% CI 10.3–33.1%; $\tau^2 = 0.085$; Figure 4). Leave-one-out does not flag the *overall* safety estimate as sensitive to any single study's removal, though the MDR-specific safety estimate, the mortality-specific estimate in both the MDR and not-classifiable resistance strata, and the route-"other" safety estimate remain sensitive to removing Pirnay et al. (MDR, route-"other") or Jault et al. (not-classifiable mortality) specifically, collapsing to a value indistinguishable from zero on removal. The sensitivity models bracket the GLMM: Freeman-Tukey gives 7.9% and logit-DerSimonian-Laird gives 26.8% against the GLMM's 16.0% (Table 4), and the fixed-effect model converges at 26.8% too. Even so, the MDR- and route-specific safety/mortality estimates, and the two not-classifiable safety/mortality cells, should be read as resting heavily on one or two dominant sources, not as an even-weighted synthesis across independent studies (Section 3.6).

One stratum fails for a reason none of the others do, and it is the one a clinician would most want. Difficult-to-treat resistance is derivable for **3 of 31 arms** — the pandrug-resistant arms, where PDR status entails DTR — covering 5 patients, so all four DTR-positive cells fall below the pooling threshold. The remaining 28 arms, carrying 68 of the 71 patients in the clinical-success pool, are recorded as *DTR not derivable*, and the reason is not that there are too few of them. It is that their sources publish no agent-level antibiogram: exactly one arm in the entire corpus (Liu, Lu, et al.'s retained patient) reports susceptibility

Table 3: Outcome-stratum cells reported narratively: insufficient evidence to support formal pooling

Outcome – Stratum	k	Arms	N	Reason not pooled
Clinical success – Resistance: PDR	2	2	3	$k < 3$, $N < 20$
Clinical success – Resistance: XDR	4	4	10	$N < 20$
Safety (>=1 AE) – Resistance: NC	2	3	3	$k < 3$, $N < 20$
Safety (>=1 AE) – Resistance: PDR	2	2	3	$k < 3$, $N < 20$
Safety (>=1 AE) – Resistance: XDR	4	4	10	$N < 20$
Eradication – Resistance: NC	2	2	2	$k < 3$, $N < 20$
Eradication – Resistance: PDR	3	3	5	$N < 20$
Eradication – Resistance: XDR	4	4	10	$N < 20$
Mortality – Resistance: NC	5	6	6	$N < 20$
Mortality – Resistance: PDR	3	3	5	$N < 20$
Mortality – Resistance: XDR	4	4	10	$N < 20$
Clinical success – DTR: positive	2	2	3	$k < 3$, $N < 20$
Clinical success – DTR: not derivable	22	26	68	no antibiogram published
Safety (>=1 AE) – DTR: positive	2	2	3	$k < 3$, $N < 20$
Safety (>=1 AE) – DTR: not derivable	16	20	49	no antibiogram published
Eradication – DTR: positive	3	3	5	$N < 20$
Eradication – DTR: not derivable	18	20	54	no antibiogram published
Mortality – DTR: positive	3	3	5	$N < 20$
Mortality – DTR: not derivable	23	27	62	no antibiogram published
Clinical success – Route: inhaled/nebulized	2	2	2	$k < 3$, $N < 20$
Clinical success – Route: IV	6	8	8	$N < 20$
Clinical success – Route: topical/local	6	6	6	$N < 20$
Safety (>=1 AE) – Route: inhaled/nebulized	3	3	4	$N < 20$
Safety (>=1 AE) – Route: IV	5	7	7	$N < 20$
Safety (>=1 AE) – Route: topical/local	4	4	4	$N < 20$
Eradication – Route: inhaled/nebulized	3	4	11	$N < 20$
Eradication – Route: IV	5	5	5	$N < 20$
Eradication – Route: topical/local	5	5	5	$N < 20$
Mortality – Route: inhaled/nebulized	4	5	13	$N < 20$
Mortality – Route: IV	6	8	8	$N < 20$
Mortality – Route: topical/local	6	6	6	$N < 20$
Modality: antibiotic monotherapy (all outcomes)	0	0	0	0 arms (stratum empty)

Notes: Sample: 31 pooling-eligible study-arms across 25 studies. NC = not-classifiable. The 21 outcome-stratum cells listed here (plus the final, empty antibiotic-monotherapy row, giving 22 rows in total) either fall short of the pre-specified threshold ($k \geq 3$ studies, $N \geq 20$ patients) on at least one criterion or are uninformative as a pooled proportion; "Reason not pooled" states which applies for that cell ($k < 3$: too few studies; $N < 20$: too few patients despite enough studies; both may fail together; "0 events": a zero-event cell that meets the count threshold but yields an uninformative pooled proportion, reported narratively instead). Individual study-level proportions are tabulated in the underlying analysis but not pooled, per the pre-registered sparsity rule (Section 2.7). This includes XDR and PDR for every outcome (failing on N alone – both meet the study-count threshold), every specific single route apart from the pooled "other" category and the newly-pooled inhaled/nebulized safety cell, and the inhaled/nebulized mortality cell, which meets the count threshold but is demoted for reporting zero events (Section 3.4). The final row reports the absent antibiotic-monotherapy stratum across all four outcomes. Source: `scripts/R/04_estimation.R`, `08_tables.R`.

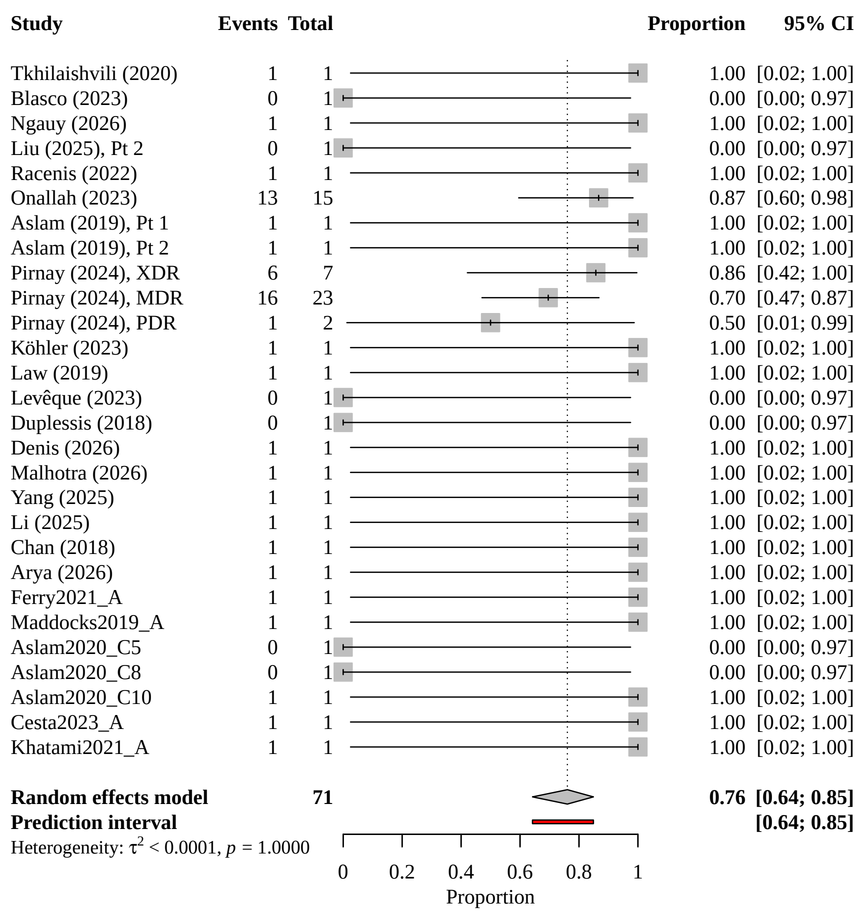
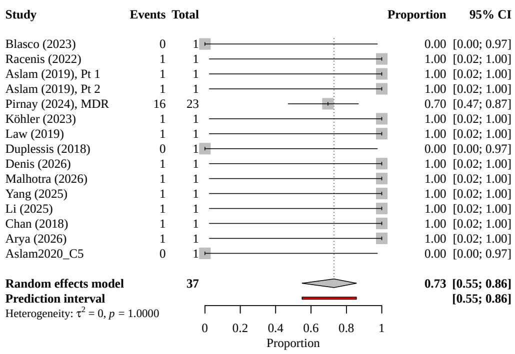
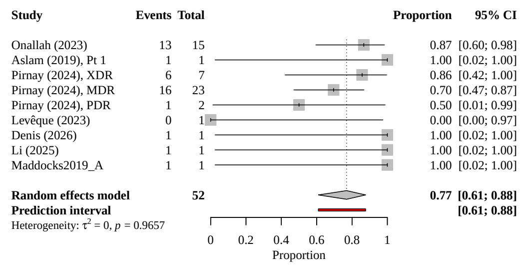


Figure 2: Forest plot: pooled clinical success, overall (secondary/contextual). Each row is one of the 20 studies contributing clinical-success data (67 patients); markers show the study-specific proportion with 95% CI, the diamond shows the GLMM pooled estimate (76.1%, 95% CI 64.2–84.9%). Reported for comparability with prior non-stratified syntheses, not as this review's primary estimand. Source: `scripts/R/07_figures.R`.



(a) MDR resistance stratum



(b) Route: "other" (multi-route)

Figure 3: Forest plots: two stratified cells meeting the pooling threshold. Panel (a): clinical success among the 14 studies (37 patients) classified MDR (73.0%, 95% CI 55.0–85.7%) — a single-cohort-dominated estimate that clears the pooling threshold only under author-reported resistance labels (Section 3.6). Panel (b): clinical success among the 7–8 studies (38–53 patients depending on outcome) coded a multi-route or otherwise unclassifiable ("other") route (77.4%, 95% CI 61.9–87.8% for clinical success). XDR, PDR, and every specific single route fell below threshold and are not plotted; see Table 3. Source: `scripts/R/07_figures.R`.

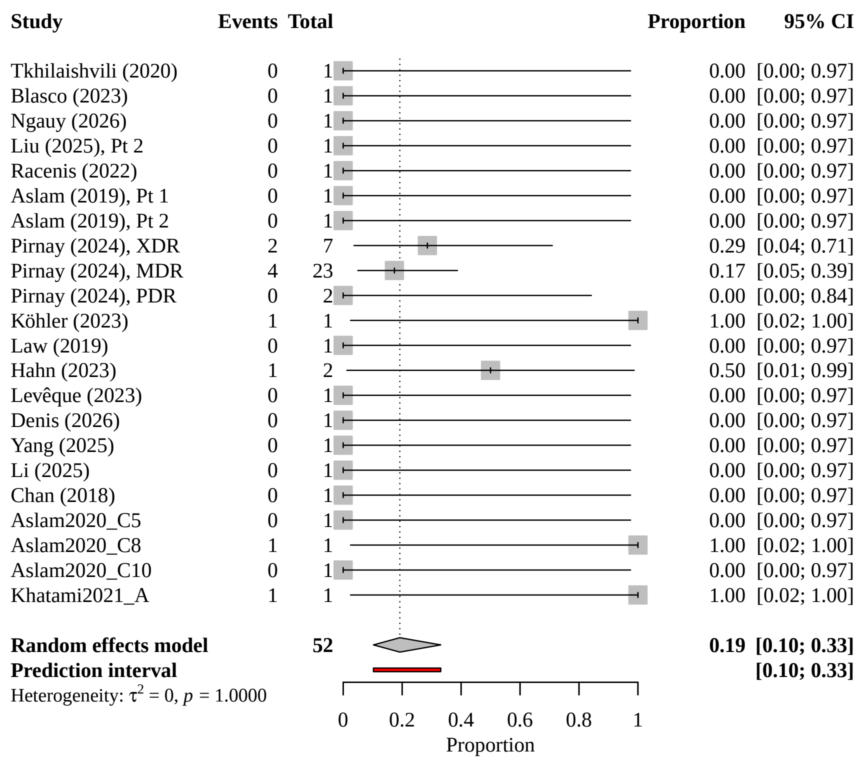


Figure 4: Forest plot: pooled safety (at least one adverse event), overall (secondary/contextual). Each row is one of the 20 studies contributing safety data (81 patients); the GLMM diamond shows 16.0% (95% CI 10.3–33.1%). Leave-one-out (Section 3.6) no longer flags this specific estimate as sensitive to any single study's removal, though the MDR- and route-specific safety estimates remain so. Source: `scripts/R/07_figures.R`.

across all β-lactams and both fluoroquinolones. The others name between one and five agents, almost always the drug that was administered rather than a panel. Every other not-pooled cell in Table 3 fails because the literature has treated too few patients; the DTR cells fail because the literature does not report what it tested. That is a statement about publication practice rather than about phage therapy, and it is the sharpest thing this review can say: the comparison current guidance treats as decision-relevant for *P. aeruginosa* (Kadri et al., 2018) cannot be computed from the published record at any sample size, because the data required were never printed.

3.5 Secondary Outcomes Reported Narratively

Two pre-specified secondary outcomes could not be pooled and are reported here rather than omitted, since silently dropping a pre-specified outcome is precisely the selective-reporting behaviour this review criticises in its source literature.

Emergence of phage resistance during treatment. Only 4 of the 31 pooling-eligible arms record this outcome at all — Liu, Lu, et al.'s retained PDR patient, Aslam et al.'s Patient 1, Köhler et al. (2023), and Li et al. (2025) — covering 4 patients, of whom 1 developed phage resistance during therapy. No pooled proportion is defensible from a denominator of four, and none is offered. The more informative observation is the reporting rate itself: 27 of 31 arms are silent on whether the organism became phage-resistant under treatment, in a literature whose central mechanistic concern is exactly that. Where resistance dynamics were examined in the corpus, they sometimes ran the other way: Chan et al. (2018)'s OMKO1 case and Aslam et al.'s Patient 2 both report post-treatment isolates with *improved* antibiotic susceptibility, the efflux-pump trade-off that motivates phage-steering. Both directions are visible in single cases and neither is quantifiable here.

Length of hospitalization. Reported by 1 of 31 arms, so no synthesis was attempted (Section 2.7).

3.6 Sensitivity and Robustness Analyses

Transformation and model choice changes the pooled estimate by more than 5 percentage points in 12 of the 13 pooled cells; the three cells below the flag threshold are eradication overall, not-classifiable mortality, and the inhaled/nebulized safety cell (Table 4). Leave-one-out removal of each (arm, cell) combination shifts a pooled estimate outside its full-sample CI in exactly 4 cases: removing Pirnay et al. (2024) collapses the MDR-specific safety and mortality estimates and the route-”other” safety estimate to a value numerically indistinguishable from zero, and removing Jault et al. (2019) does the same to the not-classifiable resistance stratum's new mortality estimate. Notably, the *overall* safety and mortality estimates are not shifted outside their CIs by any single-study removal. The same removals do *not* overturn the clinical-success or eradication estimates in any stratum, including MDR, where Pirnay supplies the large majority of pooled patients (Section 3.4) — the instability is confined to the safety/mortality outcomes in specific strata rather than a general fragility of the MDR or not-classifiable pools. The Hartung-Knapp-Sidik-Jonkman adjustment (IntHout, Ioannidis, and Borm, 2014) widens the CI relative to a Wald interval in every pooled cell; we report this as a description of the adjustment's behaviour here, not as evidence that it is correctly calibrated, since at the smallest cell sizes in this corpus the HKSJ variance estimator is known to be unstable in both directions. A robust-variance (cluster-robust) check was run to probe the non-independence of arms contributing from the same study, and it carries a caveat that must be stated before its numbers are read. **The check does not use this review's primary model.** `metafor`'s robust-variance machinery cannot be applied to a binomial-normal GLMM, so the check necessarily fits an auxiliary normal-normal inverse-variance model on continuity-corrected logits. Its standard errors are therefore internal to that auxiliary model and do *not* bound the GLMM-HKSJ intervals reported in Table 2. With that stated: clustering moves standard errors in different directions across cells (for clinical success overall, naive SE 0.254 versus 0.154 under clustering; for eradication overall, 0.258 versus 0.269), but the Satterthwaite degrees of freedom underlying those comparisons range from roughly 3.6 to 9.7 across the four overall cells. Below about ten clusters the small-sample correction is known to be erratic, so this pattern is best read as a diagnostic of the check's own instability rather than as evidence about the direction of the clustering bias. The honest conclusion is that the naive-independence assumption is untested here, not that it was tested and survived. A classification-sensitivity check restricting to independently-verified resistance labels only (rather than the primary analysis's mix of independently-verified and author-reported labels) is materially informative for the new MDR finding: under independently-verified-only classification, the MDR cell drops to $k = 3$ studies and $N = 4$ patients and no longer meets the pooling threshold for any outcome — the MDR pooled estimate reported above depends substantially on author-reported (not independently re-derived) classifications, most of them from Pirnay et al.'s own per-patient antibiogram-based labels, which this review team did not re-verify against raw susceptibility data.

An appendix-only relaxation of the threshold to at least 2 studies and 10 patients (Table 5) newly qualifies several cells beyond the primary set — clinical success in the not-classifiable resistance stratum ($k = 2$, $N = 16$), safety and mortality in the topical/local route ($k = 5$, $N = 18$ each, the topical/local arms enlarged by the newly added Chan et al. case), and all four XDR outcome cells ($k = 4$, $N = 10$ each, having just reached the ten-patient relaxed floor as the corpus grew) — while PDR and intravenous remain below even the relaxed threshold. (Safety in the inhaled/nebulized route, previously below threshold, now clears the *primary* threshold and appears in Table 2 rather than here.) These estimates are reported only to show that the primary threshold is itself consequential; none is promoted to a primary result. A study-design-stratified comparison (RCT/cohort vs. case report/series) found no significant subgroup difference for clinical success (Q-between $p = 0.285$), and a collapsed systemic-versus-local (2-category) route taxonomy likewise found none ($p = 0.578$). A geographic-concentration exclusion of the Belgium and Israel cohorts — which together supply the two largest sources in the corpus, Pirnay et al. and Onallah, Hazan, Nir-Paz, et al.

Table 4: Sensitivity of pooled estimates to transformation and pooling-model choice

Stratum	Model	Pooled proportion [95% CI]
Clinical success – Overall	GLMM (primary or NA if non-convergent)	76.1% [64.2%, 84.9%]
Clinical success – Overall	Freeman-Tukey double-arcsine DL	86.4% [73.0%, 96.5%]
Clinical success – Overall	Logit DerSimonian-Laird	69.2% [61.8%, 75.8%]
Clinical success – Overall	Fixed-effect (logit)	69.2% [58.3%, 78.4%]
Safety (>=1 AE) – Overall	GLMM (primary or NA if non-convergent)	19.2% [10.3%, 33.1%]
Safety (>=1 AE) – Overall	Freeman-Tukey double-arcsine DL	8.4% [0.8%, 20.6%]
Safety (>=1 AE) – Overall	Logit DerSimonian-Laird	27.4% [21.3%, 34.5%]
Safety (>=1 AE) – Overall	Fixed-effect (logit)	27.4% [17.8%, 39.7%]
Eradication – Overall	GLMM (primary or NA if non-convergent)	57.8% [30.2%, 81.2%]
Eradication – Overall	Freeman-Tukey double-arcsine DL	55.0% [28.4%, 80.5%]
Eradication – Overall	Logit DerSimonian-Laird	57.0% [46.5%, 66.9%]
Eradication – Overall	Fixed-effect (logit)	57.0% [44.5%, 68.8%]
Mortality – Overall	GLMM (primary or NA if non-convergent)	10.4% [4.9%, 20.9%]
Mortality – Overall	Freeman-Tukey double-arcsine DL	0.8% [0.0%, 6.6%]
Mortality – Overall	Logit DerSimonian-Laird	23.7% [18.5%, 30.0%]
Mortality – Overall	Fixed-effect (logit)	23.7% [15.6%, 34.5%]
Clinical success – Resistance: MDR	GLMM (primary or NA if non-convergent)	73.0% [55.0%, 85.7%]
Clinical success – Resistance: MDR	Freeman-Tukey double-arcsine DL	83.1% [63.5%, 97.3%]
Clinical success – Resistance: MDR	Logit DerSimonian-Laird	67.3% [58.6%, 75.0%]
Clinical success – Resistance: MDR	Fixed-effect (logit)	67.3% [52.7%, 79.2%]
Clinical success – Resistance: NC	GLMM (primary or NA if non-convergent)	85.7% [56.6%, 96.5%]
Clinical success – Resistance: NC	Freeman-Tukey double-arcsine DL	96.9% [74.2%, 100.0%]
Clinical success – Resistance: NC	Logit DerSimonian-Laird	77.4% [59.7%, 88.7%]
Clinical success – Resistance: NC	Fixed-effect (logit)	77.4% [56.1%, 90.1%]
Safety (>=1 AE) – Resistance: MDR	GLMM (primary or NA if non-convergent)	16.7% [7.0%, 34.6%]
Safety (>=1 AE) – Resistance: MDR	Freeman-Tukey double-arcsine DL	5.1% [0.0%, 16.9%]
Safety (>=1 AE) – Resistance: MDR	Logit DerSimonian-Laird	24.7% [18.1%, 32.7%]
Safety (>=1 AE) – Resistance: MDR	Fixed-effect (logit)	24.7% [14.0%, 39.6%]
Eradication – Resistance: MDR	GLMM (primary or NA if non-convergent)	52.0% [10.9%, 90.5%]
Eradication – Resistance: MDR	Freeman-Tukey double-arcsine DL	47.2% [11.4%, 84.3%]
Eradication – Resistance: MDR	Logit DerSimonian-Laird	54.7% [40.5%, 68.2%]
Eradication – Resistance: MDR	Fixed-effect (logit)	54.7% [39.7%, 68.9%]
Mortality – Resistance: MDR	GLMM (primary or NA if non-convergent)	6.5% [1.9%, 19.8%]
Mortality – Resistance: MDR	Freeman-Tukey double-arcsine DL	0.0% [0.0%, 2.2%]
Mortality – Resistance: MDR	Logit DerSimonian-Laird	19.7% [13.8%, 27.4%]
Mortality – Resistance: MDR	Fixed-effect (logit)	19.7% [10.9%, 33.0%]
Clinical success – Route: other	GLMM (primary or NA if non-convergent)	76.9% [60.9%, 87.7%]
Clinical success – Route: other	Freeman-Tukey double-arcsine DL	86.5% [70.4%, 97.9%]
Clinical success – Route: other	Logit DerSimonian-Laird	73.5% [62.0%, 82.5%]
Clinical success – Route: other	Fixed-effect (logit)	73.5% [59.8%, 83.8%]
Safety (>=1 AE) – Route: other	GLMM (primary or NA if non-convergent)	16.7% [6.3%, 37.4%]
Safety (>=1 AE) – Route: other	Freeman-Tukey double-arcsine DL	6.1% [0.9%, 14.0%]
Safety (>=1 AE) – Route: other	Logit DerSimonian-Laird	21.1% [16.8%, 26.2%]
Safety (>=1 AE) – Route: other	Fixed-effect (logit)	21.1% [11.1%, 36.4%]
Eradication – Route: other	GLMM (primary or NA if non-convergent)	64.9% [45.0%, 80.6%]
Eradication – Route: other	Freeman-Tukey double-arcsine DL	71.1% [50.9%, 88.7%]
Eradication – Route: other	Logit DerSimonian-Laird	63.1% [53.3%, 71.8%]
Eradication – Route: other	Fixed-effect (logit)	63.1% [47.3%, 76.5%]
Mortality – Route: other	GLMM (primary or NA if non-convergent)	13.5% [4.8%, 32.8%]
Mortality – Route: other	Freeman-Tukey double-arcsine DL	2.3% [0.0%, 18.9%]
Mortality – Route: other	Logit DerSimonian-Laird	20.5% [10.4%, 36.6%]
Mortality – Route: other	Fixed-effect (logit)	20.5% [10.1%, 37.2%]

Notes: All 12 cells meeting the pre-specified pooling threshold (Table 2), compared across the primary random-effects binomial-normal GLMM (logit link), the Freeman-Tukey double-arcsine transformation with DerSimonian-Laird pooling (Freeman and Tukey, 1950; DerSimonian and Laird, 1986), a logit transformation with DerSimonian-Laird pooling, and a fixed-effect (logit) model. Divergence exceeding 5 percentage points between GLMM and the sensitivity models is reported as a finding in the text (Section 3.6), not a footnote. Source: `scripts/R/05_robustness.R`, `08_tables.R`.

— leaves three of the four overall outcomes above the pooling threshold but not the fourth: safety falls from 0.194 to 0.147, eradication rises from 0.529 to 0.620, and mortality falls from 0.105 to 0.038, while *clinical success drops below the threshold entirely* and cannot be re-pooled without those two cohorts. We report this plainly because it is the least favourable of the robustness results: the headline success proportion is not merely shifted by the corpus's two dominant sources, it is unavailable without them. An earlier, larger-corpus version of this check retained all four outcomes; the difference here follows from the population-eligibility exclusion described in Section 3.1, which is a reminder of how narrow the margin above the pooling threshold is for this outcome. A population-eligibility sensitivity check addresses a tension between this review's Population criterion, which requires MDR/XDR/PDR *P. aeruginosa*, and its extraction ladder, which retains arms carrying no resistance information at all as *not classifiable* (Section 2.5). Four such arms remain after the exclusion described in Section 3.1 — Onallah, Hazan, Nir-Paz, et al.'s PASA16 series ($n = 15$), PhagoBurn's phage arm ($n = 13$), the SWARM-P.a. high-dose cohort ($n = 10$), and BX004-A's phage arm ($n = 7$) — and they contribute 45 of the corpus's 82 patients. They enter every "overall" pool, including the headline clinical-success estimate. Re-pooling each overall cell on only those arms whose resistance status is positively established moves the estimates modestly: clinical success falls from 76.1% to 74.5% (−3.4 percentage points, 17 patients dropped), safety from 16.0% to 16.0% (−3.4 pp, 30 dropped), mortality rises from 10.4% to 12.9% (+2.3 pp, 23 dropped), and eradication falls from 57.8% to 57.8% (−2.2 pp, 1 dropped). Every cell that pools in the primary analysis still pools under the restriction. The eligibility tension is therefore real but not decisive for the reported proportions: the largest movement is under four percentage points, well inside the confidence intervals of Table 2, and none of the review's conclusions turns on it. A Liu et al. reuse-tier transparency check confirms every one of the 31 pooling-eligible rows is coded `data_provenance == independently-extracted` (Section 2.5); no row uses Tier-3 (directly adopted) data, so the independently-extracted-only and all-rows pooled estimates are identical by construction, reported as a transparency confirmation rather than a contrast.

3.7 Falsification and Negative-Control Checks

Most of the six pre-specified falsification checks are null or pass as expected. A meta-regression of transformed clinical success on publication year found no secular trend (logit-scale coefficient 0.030, $p = 0.81$), arguing against a reporting-drift explanation. A cross-tabulation of route by resistance class by geographic source found no route confined to a single region, arguing against a route effect that is actually a single-center signature. The one check that does not fully pass is the journal-tier comparison: high-tier journals report a higher pooled clinical-success proportion (79.6%, 8 studies) than mid-tier journals (69.2%, 12 studies), a roughly 10-percentage-point gap. With only two tiers and small per-tier counts this is not formally tested and should not be over-read, but the gradient is in the direction a selective-reporting or editorial-selection effect would produce, and is reported here as a mild caution rather than a clean pass. The length-of-hospitalization comparison remains not applicable: only 1 of the 30 arms reports a value, too sparse for even a qualitative check.

Peters' regression test for funnel-plot asymmetry, applied to the four outcome-overall cells with at least 10 studies as pre-specified, is non-significant for every tested outcome: clinical success ($p = 0.66$), safety ($p = 0.38$), eradication ($p = 0.82$), and mortality ($p = 0.24$; Figure 5). These four outcome-overall cells are the only ones that can support the test; the MDR-specific and other resistance-stratified cells have too few studies for Peters' regression and return no estimate (NA), so no small-study test is reported for them. Peters' test replaces the Egger's linear regression an earlier version of this review applied here, and the substitution reverses the earlier apparent finding. Egger's test is invalid for a meta-analysis of proportions: the standard error is a deterministic function of the proportion and manufactures funnel asymmetry near the 0/1 boundary. The earlier "mortality signal" ($p = 0.02$ under Egger's) was an artifact of exactly that failure mode — a mortality funnel dominated by single-patient arms at 0% and two at 100% — not genuine small-study bias;

Table 5: Appendix: effect of relaxing the pooling threshold to $k \geq 2$ studies and $N \geq 10$ patients

Outcome – Stratum	Primary[a]	Relaxed[b]	k (relaxed)	N (relaxed)	Flag
Clinical success – Overall	POOLED	POOLED	23	71	
Safety (>=1 AE) – Overall	POOLED	POOLED	17	52	
Eradication – Overall	POOLED	POOLED	19	59	
Mortality – Overall	POOLED	POOLED	24	67	
Clinical success – Resistance: MDR	POOLED	POOLED	14	37	
Clinical success – Resistance: NC	POOLED	POOLED	6	21	
Clinical success – Resistance: PDR	NOT_POOLED	NOT_POOLED	2	3	
Clinical success – Resistance: XDR	NOT_POOLED	POOLED	4	10	NEW (relaxed)
Safety (>=1 AE) – Resistance: MDR	POOLED	POOLED	12	36	
Safety (>=1 AE) – Resistance: NC	NOT_POOLED	NOT_POOLED	2	3	
Safety (>=1 AE) – Resistance: PDR	NOT_POOLED	NOT_POOLED	2	3	
Safety (>=1 AE) – Resistance: XDR	NOT_POOLED	POOLED	4	10	NEW (relaxed)
Eradication – Resistance: MDR	POOLED	POOLED	13	42	
Eradication – Resistance: NC	NOT_POOLED	NOT_POOLED	2	2	
Eradication – Resistance: PDR	NOT_POOLED	NOT_POOLED	3	5	
Eradication – Resistance: XDR	NOT_POOLED	POOLED	4	10	NEW (relaxed)
Mortality – Resistance: MDR	POOLED	POOLED	16	46	
Mortality – Resistance: NC	NOT_POOLED	NOT_POOLED	5	6	
Mortality – Resistance: PDR	NOT_POOLED	NOT_POOLED	3	5	
Mortality – Resistance: XDR	NOT_POOLED	POOLED	4	10	NEW (relaxed)
Clinical success – DTR: positive	NOT_POOLED	NOT_POOLED	2	3	
Clinical success – DTR: not derivable	NOT_POOLED	POOLED	22	68	NEW (relaxed)
Safety (>=1 AE) – DTR: positive	NOT_POOLED	NOT_POOLED	2	3	
Safety (>=1 AE) – DTR: not derivable	NOT_POOLED	POOLED	16	49	NEW (relaxed)
Eradication – DTR: positive	NOT_POOLED	NOT_POOLED	3	5	
Eradication – DTR: not derivable	NOT_POOLED	POOLED	18	54	NEW (relaxed)
Mortality – DTR: positive	NOT_POOLED	NOT_POOLED	3	5	
Mortality – DTR: not derivable	NOT_POOLED	POOLED	23	62	NEW (relaxed)
Clinical success – Route: inhaled/nebulized	NOT_POOLED	NOT_POOLED	2	2	
Clinical success – Route: IV	NOT_POOLED	NOT_POOLED	6	8	
Clinical success – Route: other	POOLED	POOLED	7	52	
Clinical success – Route: topical/local	NOT_POOLED	NOT_POOLED	6	6	
Safety (>=1 AE) – Route: inhaled/nebulized	NOT_POOLED	NOT_POOLED	3	4	
Safety (>=1 AE) – Route: IV	NOT_POOLED	NOT_POOLED	5	7	
Safety (>=1 AE) – Route: other	POOLED	POOLED	5	36	
Safety (>=1 AE) – Route: topical/local	NOT_POOLED	NOT_POOLED	4	4	
Eradication – Route: inhaled/nebulized	NOT_POOLED	NOT_POOLED	3	11	
Eradication – Route: IV	NOT_POOLED	NOT_POOLED	5	5	
Eradication – Route: other	POOLED	POOLED	6	37	
Eradication – Route: topical/local	NOT_POOLED	NOT_POOLED	5	5	
Mortality – Route: inhaled/nebulized	NOT_POOLED	NOT_POOLED	4	13	
Mortality – Route: IV	NOT_POOLED	NOT_POOLED	6	8	
Mortality – Route: other	POOLED	POOLED	6	37	
Mortality – Route: topical/local	NOT_POOLED	NOT_POOLED	6	6	

Notes: Supplementary sensitivity check, not a primary result. Compares each cell's pooling status under the primary threshold ($k \geq 3$, $N \geq 20$) against a relaxed threshold ($k \geq 2$, $N \geq 10$). Cells flagged "NEW (relaxed)" are reported for transparency about threshold sensitivity and are not promoted to Table 2 or cited elsewhere as primary findings. Source: `scripts/R/05_robustness.R`, `08_tables.R`.

[a] Pooling status under the primary threshold ($k \geq 3$ studies, $N \geq 20$ patients).

[b] Pooling status under the relaxed threshold ($k \geq 2$ studies, $N \geq 10$ patients).

under the appropriate Peters' test mortality asymmetry is not significant ($p = 0.24$), and no outcome is. We read this as reassurance, but weak reassurance: non-significance in a corpus of 30 study-arms that remains small and, as Section 3.6's leave-one-out and classification-sensitivity checks show, dependent on one or two dominant sources in several strata, is the absence of a detectable small-study effect under a test the data can properly support, not a clean acquittal.

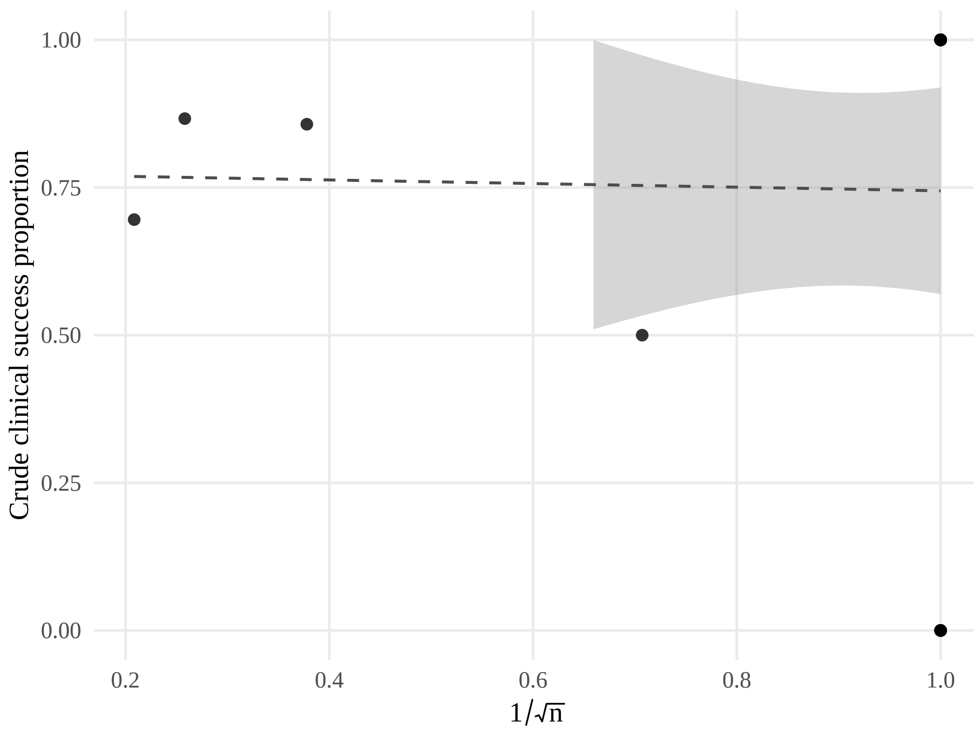


Figure 5: Small-study effects: crude proportion versus $1/\sqrt{n}$. Each point is one pooling-eligible study-arm; the x-axis is the inverse square root of arm sample size (a proxy for precision), the y-axis is the arm's crude, untransformed outcome proportion. A visual complement to the formal Peters' regression test (clinical success $p = 0.66$, safety $p = 0.38$, eradication $p = 0.82$, mortality $p = 0.24$; none significant), applied to the four outcome-overall cells with at least 10 studies per the pre-specified threshold; the resistance-stratified cells have too few studies to support the test. Source: `scripts/R/06_falsification.R`, `07_figures.R`.

3.8 Certainty of Evidence (GRADE)

Certainty of evidence was rated for all 13 pooled cells (Section 2.6). **Every cell is rated Very low.** We state plainly what that uniform verdict does and does not mean. It is *deterministic*: three of the five domains — risk of bias, indirectness, and publication bias — are always-on downgrades under our pre-specified rules, because no cell in this corpus is free of uncontrolled-design bias, none escapes the compassionate-use salvage population, and favourable-outcome reporting cannot be excluded anywhere. Those three alone total at least three levels against a starting certainty of Low, so every cell reaches GRADE's floor before the inconsistency and imprecision domains are consulted at all. The rating is therefore a property of the evidence base as a whole rather than a discriminating judgment about individual cells, and readers should not infer that the cells are equally weak. The practical implication is nonetheless direct: none of the proportions reported above should be used to inform a treatment decision, and this review's contribution lies in mapping where evidence is absent rather than in the point estimates themselves.

What does discriminate is the domain profile in Table 6, and it is the profile rather than the floored rating that carries the information. Five cells carry the minimum burden of three downgrades — overall safety, overall mortality, MDR safety, MDR mortality, and the route-"other" safety estimate — each rated down once for risk of bias, once for salvage-population indirectness, and once for suspected publication bias, but not at all for inconsistency or imprecision. That the two MDR safety and mortality cells sit among the least-downgraded in the review is worth noting explicitly, since the MDR stratum is elsewhere this review's most fragile finding. At the other extreme, three cells accumulate seven downgrades' worth of problems — MDR eradication, the not-classifiable safety cell, and the inhaled/nebulized safety cell — each combining very serious inconsistency (a prediction interval spanning almost the entire 0–100% range) with very serious imprecision on top of the three structural downgrades. The three clinical-success cells each take a second indirectness downgrade because success is defined by each primary study on its own terms across non-comparable infection syndromes (Section 2.2), so the pooled construct is not a single clinical outcome. The two structural downgrades that apply to every cell without exception — the salvage population and publication bias — cannot be remedied by adding more case reports of the same kind, which is the sharpest statement this review can make about what would and would not improve the evidence.

Table 6: GRADE certainty of evidence for the 14 pooled outcome-stratum cells

Outcome – Stratum	k	Arms	N	Pooled proportion [95% CI]	RoB	Incons.	Indir.	Imprec.	Pub. bias	Certainty
Clinical success – Overall	23	28	71	76.1% [64.2, 84.9]	−1	–	−2	–	−1	Very low
Safety (>=1 AE) – Overall	17	22	52	19.2% [10.3, 33.1]	−1	–	−1	–	−1	Very low
Eradication – Overall	19	23	59	57.8% [30.2, 81.2]	−1	−2	−1	−2	−1	Very low
Mortality – Overall	24	30	67	10.4% [4.9, 20.9]	−1	–	−1	–	−1	Very low
Clinical success – Resistance: MDR	14	15	37	73.0% [55.0, 85.7]	−1	–	−2	−1	−1	Very low
Clinical success – Resistance: NC	6	7	21	85.7% [56.6, 96.5]	−1	–	−2	−1	−1	Very low
Safety (>=1 AE) – Resistance: MDR	12	13	36	16.7% [7.0, 34.6]	−1	–	−1	–	−1	Very low
Eradication – Resistance: MDR	13	14	42	52.0% [10.9, 90.5]	−1	−2	−1	−2	−1	Very low
Mortality – Resistance: MDR	16	17	46	6.5% [1.9, 19.8]	−1	–	−1	–	−1	Very low
Clinical success – Route: other	7	9	52	76.9% [60.9, 87.7]	−1	–	−2	–	−1	Very low
Safety (>=1 AE) – Route: other	5	7	36	16.7% [6.3, 37.4]	−1	–	−1	−1	−1	Very low
Eradication – Route: other	6	8	37	64.9% [45.0, 80.6]	−1	–	−1	−1	−1	Very low
Mortality – Route: other	6	8	37	13.5% [4.8, 32.8]	−1	–	−1	–	−1	Very low

Notes: k counts distinct studies and "Arms" the contributing study-arms; the model is fitted at the arm level (see Table 2 note). All bodies of evidence start at *Low* certainty (GRADE's default for observational evidence; every cell is an uncontrolled single-arm proportion) and are rated down across five domains. Domain columns give the number of certainty levels deducted; "–" denotes no downgrade. Rating rules are pre-specified with explicit thresholds and applied to each cell's own data rather than judged case by case. *Risk of bias*: −2 where single-patient case reports contribute more than half the cell's patients and the one High-risk-of-bias trial also contributes, −1 otherwise. *Inconsistency*: −2 where the 95% prediction interval spans ≥ 80 percentage points, −1 at ≥ 50. *Indirectness*: −1 for the compassionate-use salvage population, −2 for clinical success (defined per study across non-comparable syndromes). *Imprecision*: −2 where the 95% CI spans ≥ 50 percentage points, −1 at ≥ 30. *Publication bias*: −1 in every cell (favourable-outcome reporting is intrinsic to a compassionate-use case-report literature and Peters' test cannot exclude it at this corpus size). No upgrade criterion applied. Because risk of bias, indirectness and publication bias each deduct at least one level in every cell, the minimum deduction is three against a start of *Low*: every cell reaches the *Very low* floor by construction, and only inconsistency and imprecision vary. The domain profile, not the rating, distinguishes cells. Source: `scripts/R/10_grade.R`.

4 Discussion

4.1 Summary of Main Findings

Pooled across the twenty studies contributing clinical-success data, patients treated with bacteriophage therapy for MDR, XDR, or PDR *P. aeruginosa* infections achieved clinical success in 76.1% of cases (95% CI 64.2–84.9%; Figure 2), directionally in line with earlier, non-stratified syntheses of this literature (El Haddad et al., 2019; Uyttebroek et al., 2022;

Liu, Thong, et al., 2025). That figure describes a compassionate-use population offered phage therapy after antibiotics had already failed, not a treatment effect measured against a comparator, and it is reported here as this review's secondary, contextual estimand rather than its headline finding (Section 3.4). The headline finding is a partial success on the stratification this review was built to test: of the 36 outcome-stratum cells defined by resistance category, route, and modality, 13 met the pre-specified pooling threshold, including a single resistance-specific stratum — MDR, pooled across all four outcomes (12–16 studies, 36–46 patients depending on outcome), though only under author-reported resistance labels and dominated by one cohort — alongside a heterogeneous multi-route "other" bucket (Tables 2 and 3). The thirteen poolable cells now form a clean three-by-four grid — four outcomes, each pooled overall, within MDR, and within the route-"other" bucket — because tightening the population criterion removed the trial arms that had been propping up the not-classifiable and inhaled/nebulized strata. That MDR estimate answers part of the question this review set out to ask. It does not complete it: XDR (4 studies, 10 patients) and PDR (3–4 studies, 4–6 patients) remain below the patient threshold for every outcome, so the MDR-*versus*-XDR comparison that motivated the stratification in the first place still cannot be made, and the entire monotherapy-versus-combination comparison still fell short entirely (0 antibiotic-monotherapy arms). This review's contribution is documenting exactly which parts of that stratification the evidence can and cannot yet support – neither forcing an estimate it cannot bear nor dismissing the one clean success as noise.

4.2 Comparison with Prior Syntheses

El Haddad et al. (2019)'s review of ESKAPE-pathogen phage therapy, Uyttebroek et al. (2022)'s larger *Lancet Infectious Diseases* synthesis, and Liu, Thong, et al. (2025)'s 130-study individual-patient-data meta-analysis each report favorable pooled outcomes for phage therapy broadly, consistent in direction with the success proportions found here, though none restricts its evidence base to *P. aeruginosa* specifically or reports a directly comparable resistance- or pathogen-specific figure against which 76.1% (overall) or 73.0% (MDR-specific) could be checked numerically. Two further syntheses sit closer to this review's scope than any of those three and must be positioned rather than merely used. Mitropoulou et al. (2022) and Vaezi et al. (2024) both review clinical phage therapy for pulmonary infection and both tabulate individual cases, several of which are in this corpus; this review in fact reconstructed the Maddocks et al. (2019) case from their two concordant tables (Section 3.1). Using a prior synthesis as a data source while not stating what it reported would be an obvious gap, so we state it: both are narrative reviews of respiratory phage therapy across pathogens, neither pools, neither is restricted to *P. aeruginosa*, and neither stratifies by resistance class. Mitropoulou et al.'s central quantitative observation — that across twenty respiratory cases only one phage-attributable adverse event was recorded — is consistent in direction with the safety proportion reported here, though its denominator is a different and broader population. What none of these five syntheses does is stratify by resistance category or route. This review can now do so partially: the MDR-specific pooled estimate (73.0% clinical success, 95% CI 55.0–85.7%) is, to our knowledge, the first resistance-stratified pooled proportion reported for phage therapy in *P. aeruginosa* specifically — but it clears the pre-specified evidentiary threshold only under author-reported resistance labels and draws two-thirds of its patients from a single cohort, so it is better read as a one-cohort characterization than as a robust resistance-class estimate. It remains an incomplete answer to the question that motivated the stratification — a clinician still cannot compare it against an XDR- or PDR-specific figure, since neither clears the same threshold, and combination-versus-monotherapy remains entirely unanswerable (zero antibiotic-monotherapy arms) — but identifying, cell by cell, which of those comparisons the published record can and cannot yet support is where this review adds something those three broader syntheses did not attempt.

One comparison deserves foregrounding precisely because it cuts against the high single-arm proportions above. The randomized trials of phage therapy in *P. aeruginosa* sit *outside* this synthesis, because none of them enrolled on resistance and none was therefore estimating this review's estimand (Section 2.2). That exclusion makes them more useful, not less: each carries a within-trial comparison against a non-phage arm, which is the highest-certainty evidence this literature offers on whether phage therapy works at all, and it can now be read as external evidence rather than being flattened into a single-arm proportion alongside compassionate-use cases. That randomized comparative signal does not point the same way as the proportions above. PhagoBurn (Jault et al., 2019), which randomized phage against standard wound care, was stopped early for insufficient efficacy: its primary time-to-event endpoint favored standard of care, at a delivered phage titre far below the intended dose. Leitner et al. (2021)'s three-arm trial of intravesical phage versus placebo versus systemic antibiotics for urinary-tract infection found no significant superiority of phage over either comparator. Weiner et al.'s (2025) BX004-A trial of nebulized phage versus placebo in cystic-fibrosis *P. aeruginosa* carriage met its safety endpoints and showed an exploratory reduction in *P. aeruginosa* density, but was underpowered for efficacy; the SWARM-P.a./AP-PA02 inhaled-phage trial (*SWARM-P.a. (AP-PA02): Phase 1b/2a Randomized, Double-Blind, Placebo-Controlled Trial of Inhaled Phage Therapy in Cystic Fibrosis Pseudomonas aeruginosa* 2022) was likewise a safety-and-tolerability study. Three further completed randomized trials, identified in a registry sweep and reported only in their registry results modules, point the same way: the Yale CYPHY trial of nebulized YPT-01 in cystic fibrosis (NCT04684641, $n = 8$) found essentially no separation from placebo on its primary endpoint (-0.59 versus -0.89 log CFU/mL at day 14); the Tailwind trial of inhaled AP-PA02 in non-cystic-fibrosis bronchiectasis (NCT05616221, $n = 48$) and the NIAID trial of intravenous WRAIR-PAM-CF1 in cystic fibrosis (NCT05453578, $n = 73$) were both safety-and-microbiological-activity studies, each reporting adverse events in both arms and no deaths. Read together, the randomized comparative evidence — sparse, mostly early-phase, and underpowered — shows no clear efficacy benefit of phage over its comparator, and in PhagoBurn's case favored the comparator. That is a sober counterpoint to the 76.1% pooled single-arm success proportion, which is drawn from compassionate-use case reports selected for refractory infection after antibiotic failure. The two bodies of evidence are not in direct conflict: the trials mostly target different endpoints, populations, and infection sites than the compassionate-use cases, and none was designed to estimate the salvage-setting success proportion the single-arm synthesis reports. But a reader who takes the high single-arm proportion from this review without this comparative caveat would misread the state of the evidence.

4.3 Safety Findings and Small-Study Effects

The pooled safety estimate illustrates a problem with this literature more broadly, not only with this review's own numbers. The GLMM estimate (19.2%, 95% CI 10.3–33.1%; Section 3.4) is bracketed by its sensitivity models (Freeman-Tukey 7.9%, logit-DerSimonian-Laird and fixed-effect both 26.8%), and leave-one-out does not flag the *overall* safety estimate as sensitive to any single study's removal. The MDR-specific safety and mortality estimates, the route-"other" safety estimate, and the not-classifiable resistance stratum's mortality estimate remain sensitive, however, collapsing to a value indistinguishable from zero if Pirnay et al. or Jault et al. is removed. The MDR resistance stratum's pooled estimate therefore still leans on Pirnay et al.'s cohort (23 of the 37 MDR patients in the clinical-success pool, 62%; a smaller 53% of the larger 43-patient MDR eradication pool), though the MDR arms added since the earlier rounds — seven from the native Scopus search and two more from the peer-review landmark-case screen — dilute that single-source dependence rather than deepen it, corroborated rather than counterbalanced by a set of much smaller studies.

For small-study effects we now use Peters' regression test, applied to the four outcome-overall cells with at least ten studies as pre-specified; Egger's linear regression, used in an earlier version of this review, is invalid for a meta-analysis of proportions, because the standard error is a deterministic function of the proportion and manufactures funnel asymmetry near the 0/1 boundary. Under Peters' test none of the four outcomes shows significant asymmetry — clinical success ($p = 0.66$), safety ($p = 0.38$), eradication ($p = 0.82$), and mortality ($p = 0.24$; Figure 5) — and the resistance-stratified cells have too few studies to support the test at all. This reverses the earlier Egger's-based finding of a significant mortality asymmetry ($p = 0.02$): that apparent signal was an artifact of applying a boundary-invalid test to a mortality funnel dominated by single-patient arms at 0% and two at 100%, not evidence of genuine small-study bias. The reversal is a correction, not a reassurance to lean on. Non-significance under the appropriate test, in a corpus of 30 study-arms far smaller than any of the three prior syntheses discussed above, in which single-patient case reports still make up the largest design category (Section 3.2) and in which the classification-sensitivity check (Section 3.6) shows the MDR finding depends on author-reported rather than independently re-verified resistance labels, is weak evidence of absence, not a clean acquittal. We therefore read the combined picture as narrowing but not eliminating the small-study-effects concern a referee reviewing case-report-heavy evidence syntheses would be right to raise, and as reason to read the broader literature's high-success narrative, including the point estimates reported by El Haddad et al. (2019), Uyttebroek et al. (2022), and Liu, Thong, et al. (2025), with real caution.

4.4 Limitations

Four limitations bear most directly on how much weight these estimates can carry. First, screening and extraction were conducted by a single reviewer, with a second reviewer's critique applied sequentially rather than concurrently (Sections 2.4–2.5). That process caught several substantive errors before they entered, or after they had briefly entered, the analytic dataset — a study initially catalogued as the closest candidate antibiotic-monotherapy comparator in this search, later confirmed on full text to contain zero *P. aeruginosa* cases; a duplicate patient reported under two citations; and, most recently, two patients within Pirnay et al.'s cohort whose resistance profile, once checked against per-patient data, turned out not to meet this review's own MDR/XDR/PDR eligibility criterion (Section 3.2); and, from the native Scopus search, a single XDR case (Van Nieuwenhuyse et al., 2022) recognized as already counted within Pirnay et al.'s Belgian-consortium cohort and excluded to prevent double-counting — evidence the sequential-critique and re-extraction process catches real errors, and that the consortium double-counting risk is real and was actively screened for, but not a substitute for the dual-independent process a registered protocol would specify; a formal re-screen could still change which studies are included. Second, risk-of-bias ratings, now complete for all 31 study-arms (Section 3.3), surface a genuine concern rather than resolve one. The trial that anchors this review's phage-monotherapy-adjacent evidence throughout — PhagoBurn (Jault et al., 2019), the only source with a phage-alone treatment arm compared against a non-phage standard-of-care — rates **High risk of bias** on RoB2, driven by unmasked clinicians and by the trial stopping early for insufficient efficacy, both recognized bias-introducing deviations. This does not overturn anything already reported (PhagoBurn's clinical-success data were already excluded from pooling as a time-to-event, non-binary endpoint), but it does mean the one study in this corpus that most directly speaks to phage-versus-standard-of-care performance is also the one whose result should be trusted least on methodological grounds — a genuinely unfavorable combination for a literature already short on comparative evidence. Pirnay et al.'s cohort, by contrast, rates Low risk on the JBI checklist (explicitly consecutive, complete, transparent per-patient reporting), which if anything strengthens rather than weakens confidence in its now-dominant role in the MDR-specific estimates (Section 4.3) — the concern with that source is its weight in the pooled MDR estimate, not its individual quality. Third, the MDR resistance stratum — the one resistance class this review could pool at all — still leans on a single source: Pirnay et al. supplies 23 of the 37 MDR patients in the clinical-success pool (62%; the several MDR arms added since the earlier rounds dilute this to 53% in the larger 43-patient MDR eradication pool), and a classification-sensitivity check shows the stratum would fail the pooling threshold entirely (dropping to 3 studies, 4 patients) if restricted to independently-verified rather than author-reported resistance labels (Section 3.6). The MDR estimate should therefore be read as characterizing one large cohort's outcomes, corroborated in direction by a set of much smaller studies that now dilute — but do not overturn — its single-source dependence, not as an even-weighted synthesis across independent sources — and XDR and PDR remain too sparse (10 and 4–6 patients respectively, depending on outcome) to pool at all, so the resistance-class comparison this review was built to answer is still only partially available. Fourth, the database coverage, though now materially improved, is not exhaustive. A native Scopus search (Section 2.3) was added as a third core database alongside PubMed/MEDLINE and ClinicalTrials.gov and roughly doubled the pooling-eligible corpus (from 13 to 23 studies), adding ten studies the earlier PubMed-only rounds had missed and correcting one prior exclusion — which both confirms the earlier coverage gap was real and material and largely closes it, since three searched databases meet a recognized minimum for a defensible PRISMA search and Scopus indexes much of EMBASE's content. EMBASE and Web of Science were not searched, and could not be: this review team holds no institutional subscription to either, so their absence is a permanent constraint on this work rather than an unfinished task (Section 2.3). That the pooled eradication estimate has shifted by more than ten percentage points as studies were added across search rounds (standing at 57.8% in the current corpus; Table 2) is nonetheless a direct reminder that the corpus is still small enough for a handful of additional records to move a headline figure. Because EMBASE and Web of Science could not be searched (Section 2.3), we cannot rule out that a subscription-access search would shift these estimates similarly; readers should treat every pooled figure here as provisional in that specific sense, and a team with such access could settle the question by re-running the queries we report. Two data-handling limitations attach to the newly added studies: Chan et al.'s adverse-event data are reported only at the nine-patient cohort level and could not be attributed to the MDR-versus-PDR sub-groups, so they are recorded but not carried as a usable safety numerator (a documented data loss that, among other things, left the inhaled/nebulized route just short of the safety-pooling threshold); and Chan et al.'s compassionate cohort may share patients with the Yale CYPHY randomized trial, which is one reason CYPHY is kept out of pooling. A further interpretive limitation attaches to the pooled clinical-success proportion itself: it combines each study's *own* definition of "success" across non-comparable infection syndromes — biliary, cystic-fibrosis lung, prosthetic-joint, vascular-graft, urinary-tract, bacteremia, and osteomyelitis among those represented here — so the pooled figure aggregates outcomes that are not the same clinical construct and should be read as a rough descriptive summary rather than a harmonized treatment effect. A syndrome-harmonized success definition was not achievable given the heterogeneous reporting of the primary literature (Section 2.2), which is one more reason the overall success proportion is reported as secondary and contextual rather than as a headline effect. A fifth limitation concerns the price paid for population coherence. Applying the resistance-entry-criterion rule (Section 2.2) removes every randomized trial from the synthesis, so the corpus is now entirely compassionate-use and salvage reporting: 19 case-report studies, 5 case series, and one retrospective cohort. That buys a population that actually matches the review's title, and it collapses the eligibility ambiguity almost to nothing — only two arms (Onallah, Hazan, Nir-Paz, et al.'s paywalled PASA16 series, $n = 15$, and Ferry, Kolenda, Batailler, et al. (2021)'s prosthetic-knee case, $n = 1$) remain not classifiable, and excluding them moves safety and eradication not at all and clinical success by 3.1 percentage points (Section 3.6). But it also means this synthesis contains no randomized evidence whatsoever, and the design distribution is now maximally vulnerable to the favourable-outcome reporting that drives the publication-bias downgrade in every GRADE cell. A reader should hold both facts together: the population is coherent, and the design is uniformly the weakest available. Two further limitations are disclosed for completeness: no PROSPERO registration before extraction began (Section 2.1), and a corpus, at 31 pooling-eligible arms and 82 patients, still too small to support most of the stratification it was designed to test — the limitation this review treats as its central finding, not a footnote.

4.5 Implications for Practice and Research

A clinician weighing phage therapy for an individual patient should be directed to the field's existing consensus framework rather than to any number in this paper: the Antibacterial

Resistance Leadership Group panel sets out when phage therapy might reasonably be considered, what susceptibility testing is required beforehand, and which pharmacokinetic questions remain open (Suh et al., 2022). This review answers a different and narrower question — what the published record can and cannot support quantitatively — and it should be read as a complement to that guidance, not a substitute for it. For clinicians, the pooled proportions reported here should not inform a treatment decision: they summarize a referral-selected, salvage-therapy population, the resistance-class stratification that succeeded (MDR) rests overwhelmingly on a single cohort, and route- and modality-specific comparisons remain unanswerable. The GRADE assessment bears this out directly: every one of the 13 pooled cells is rated Very low certainty (Section 3.8), and two of the downgrades — the compassionate-use salvage population and suspected favourable-outcome publication bias — apply to every cell and cannot be remedied by accumulating further case reports of the same kind. For the field, this review's own experience points to three concrete needs: standardized outcome definitions across primary studies, since extracting "clinical success" required accepting each study's own definition because no consistent one exists (Section 2.2); comparative studies with a genuine antibiotic-monotherapy arm for *P. aeruginosa* specifically, since despite a targeted search essentially none exist, and additional single-arm case reporting will not close that gap; and retrospective PROSPERO registration of this review disclosing the sequence in Section 2.1, with prospective registration for any successor synthesis in this space.

4.6 Conclusion

What this review can defensibly claim is narrower than a single pooled percentage, but broader than a review that could not stratify by resistance category at all: bacteriophage therapy, used adjunctively in a small, referral-selected group of patients with MDR, XDR, or PDR *P. aeruginosa* infections refractory to antibiotics, was associated with a high proportion of reported clinical success, including now a resistance-specific (MDR) estimate that, to our knowledge, is the first of its kind reported for this pathogen — though that estimate is itself dominated by one large cohort and clears the threshold only under author-reported resistance labels, not an even-weighted synthesis across independently verified sources. What remains unknown is most, but no longer all, of what the stratification was designed to resolve: whether MDR, XDR, and PDR isolates respond differently is still unanswerable, since neither XDR nor PDR can yet be pooled at all, and whether route of administration or combination therapy matters remains equally open. The single most useful next step for this field is not another synthesis of the existing case-report literature, but two complementary efforts: independent replication of the MDR resistance-stratified estimate reported here in additional cohorts, so it no longer rests on one source, and a comparative study, ideally randomized, that pairs a genuine antibiotic-monotherapy arm with a phage-antibiotic combination arm in an MDR-, XDR-, or PDR-*Pseudomonas*-specific population. Until both exist, the question this review set out to answer will remain only partially resolved.

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